

# HJ-Gauss

## Monte-Carlo Hamilton-Jacobi Reachability

**Full Exposition:** From TVD-RK Levelsets to Monte Carlo Sampling for BRTs.

**Lekan Molu · Amazon IRG**

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with V. Renganathan (Cranfield) and N. Cho (Seoul National University)

## BLUF (Bottom Line Up Front) — 1/2

Grid-resolved Hamilton-Jacobi (HJ) BRTs yield certified safety sets at a  $O(M^n)$  memory cost.

-  **Background:** LevelSetPy resolves backward reachable sets/tubes by discretizing the HJ PDE (upwinding + Lax-Friedrichs + TVD-RK) on a grid — certified, but exponential in state dimension  $n$ .
-  **Idea:** a Cole-Hopf transformation turns the *viscous* HJ PDE into a **linear heat equation**; Feynman-Kac evaluates it as a **Gaussian expectation**. No grid.

HJ-Gauss replaces the grid with Gaussian Monte-Carlo, dropping the footprint to  $O(N \cdot n)$ .

## BLUF (Bottom Line Up Front) — 2/2

HJ-Gauss replaces the grid with Gaussian Monte-Carlo, dropping the footprint to  $O(N \cdot n)$ .

-  **Generality:** a frozen-coefficient **Picard iteration** handles the nonconvex, state-dependent Hamiltonians of reachability.
-  **Rigor:**  $O(N^{-1/2})$  concentration, contraction convergence, a linearization residual bound, and a **conservative safety certificate**.
-  **For this room:** the windowed reachable tube is a dynamics-aware **conflict predicate** for MAPF; the value gradient is a **score** for diffusion planners.

This Gaussian kernel is the diffusion kernel, the bridge to learning-based planners.

## Roadmap — Main Talk (Parts 1-5)

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| Part | Theme                                                         |
|------|---------------------------------------------------------------|
| 1    | The wall: curse of dimensionality & prior grid-free work      |
| 2    | HJ-Gauss core theory: Cole-Hopf → heat → Feynman-Kac → Picard |
| 3    | Importance sampling & variance control                        |
| 4    | Guarantees: concentration, contraction, residual, certificate |
| 5    | Experiments (rockets, Dubins, 45D, $10^5$ birds)              |

 **Continue:** Parts 6-9 on the next slide.

## Roadmap — Main Talk (Parts 6-9) & Appendix

| Part                                                                                  | Theme                                                                                           |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 6                                                                                     | Application: multi-agent path finding (AMFS)                                                    |
| 7                                                                                     | The diffusion connection                                                                        |
| 8                                                                                     | Dirty Laundry: limits & boundaries                                                              |
| 9                                                                                     | Conclusions, future work, references                                                            |
|  A–C | Appendix: HJ/viscosity theory (A) · reachability foundations (B) · LevelSetPy grid pipeline (C) |

The background — **HJ PDE and viscosity theory (A)**, **Reachability Foundations (B)**, and the **LevelSetPy grid pipeline (C)** — is collected in the **Appendix**.

## Notations

| Symbol                                | Meaning                                                                               |
|---------------------------------------|---------------------------------------------------------------------------------------|
| $x \in \Omega \subseteq \mathbb{R}^n$ | $x$ : State; $\Omega$ : Open set; $n$ = State dimension.                              |
| $v(t, x)$                             | Value function; $v_t$ time derivative; $Dv = \nabla_x v$ spatial gradient (co-state). |
| $H(t; x, p)$                          | Hamiltonian; $p$ = co-state.                                                          |
| $g(x)$                                | Terminal/target datum (signed distance $\ell(x)$ ); BUC.                              |
| $\delta > 0$                          | Viscosity parameter.                                                                  |



## Notations

| Symbol                           | Meaning                                                                         |
|----------------------------------|---------------------------------------------------------------------------------|
| $\omega^\delta = e^{-cv^\delta}$ | Cole-Hopf transformed variable.                                                 |
| $c(t; x)$                        | Frozen coefficient $= \frac{2}{\delta} H^\delta /  Dv^\delta ^2$ .              |
| $M, N$                           | Grid points per dimension ( $M$ ); Monte-Carlo samples per query state ( $N$ ). |
| $\mathcal{L}_0, \mathcal{L}$     | Target set, backward reachable tube (BRT).                                      |

**Two independent counts:**  $M$  evaluation states (arbitrary, grid-free) vs  $N$  Gaussian samples drawn *per state*. Total randomness per iteration  $= M \times N$ .

# Part 1

## The Wall

Curse of dimensionality, and who has tried to climb it.

## The Curse of Dimensionality, Quantified

Grid memory is  $O(M^n)$  with  $M$  points per dimension:

| State dim $n$             | $M = 100$ cells | Feasibility (double array)                  |
|---------------------------|-----------------|---------------------------------------------|
| 2                         | $10^4$          | ✓ trivial.                                  |
| 3                         | $10^6$          | ✓ easy.                                     |
| 4                         | $10^8$          | ⚠ ~0.8 GB, heavy.                           |
| 5                         | $10^{10}$       | ✗ ~80 GB.                                   |
| 6                         | $10^{12}$       | ✗ ~8 TB.                                    |
| 45 (our multi-agent game) | $10^{90}$       | ✗ exceeds atoms in the observable universe. |

- GPU acceleration lowers wall-clock, **not** the exponent.

Reachability's most valuable regime — many interacting agents, rich dynamics — is where grids are impossible.



## Attempt 1: PINN Reachability (DeepReach)

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- **DeepReach** trains a neural network to minimize the HJ PDE residual directly (physics-informed loss), no reference grid.
- **Pros:** Trades grid storage for network weights; sidesteps  $O(M^n)$  memory.
- **Con:** Scales only to moderate dimensiond (reported ~9-10D);
  - And **accuracy degrades** as  $n$  grows; training is a nonconvex optimization with no viscosity-solution guarantee.

**Catch-22:** Learning-based solvers help, but do not deliver a certified, dimension-robust representation.  
**HJ-Gauss** keeps the viscosity-solution semantics.



## Attempt 2: Convex-Duality Formulas (Hopf / Lax-Oleinik)

HJ-Gauss trades their exactness for generality, at the quantified price of a linearization residual bound (Part 4).

- **Hopf** and **Lax-Oleinik** formulas<sup>1</sup> evaluate the value **pointwise** via convex optimization — grid-free at a point.
- **Limitation:** they require a **convex** (or state-independent) Hamiltonian.
- Reachability's  $H(x, p) = \max_u \min_w \langle p, f \rangle$  is **state-dependent and nonconvex** → outside this class.

It handles the nonconvex, state-dependent Hamiltonian head-on.

<sup>1</sup> Darbon-Osher 2016; Chow-Darbon-Osher-Yin 2017; Kirchner 2018

## Attempts 3-4: Decomposition, Path Integral, Sampling

| Family                                            | Idea                                                                              | Boundary                                                                                                                     |
|---------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| System decomposition <sup>1</sup>                 | Remove dimension by splitting self-contained subsystems                           | Only when coupling structure permits; HJ-Gauss is indifferent to coupling and <i>complements</i> it.                         |
| Stochastic PDE / path integral <sup>2</sup>       | Log-substitution linearizes the <b>stochastic</b> HJB under noise-control duality | Diffusion comes from <i>process noise</i> ; ours is a <b>deterministic worst-case game</b> , $\delta$ an analysis parameter. |
| Sample-based stochastic reachability <sup>3</sup> | Certify <b>probabilistic</b> reach-avoid                                          | We certify <b>adversarial worst-case</b> reachability — a different, stronger guarantee.                                     |

**The open gap:** a grid-free, storage-light scheme for **nonconvex, state-dependent, adversarial** Hamiltonians. That is HJ-Gauss.

<sup>1</sup> Chen-Herbert · <sup>2</sup> Kappen; Theodorou · <sup>3</sup> Summers-Lygeros; Lesser-Oishi



## Where HJ-Gauss Sits

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- Keeps the **viscosity-solution** semantics (unlike PINN heuristics).
- Handles **nonconvex, state-dependent**  $H$  (unlike Hopf/Lax-Oleinik).
- **Indifferent to coupling structure** (complements decomposition).
- Certifies **adversarial worst-case** safety (unlike probabilistic sampling).
- Memory  $O(N \cdot n)$ , grid- and discretization-free.

The rest is *how*: a Cole-Hopf transformation, a Feynman-Kac expectation, and a frozen-coefficient Picard iteration.

# Part 2

## HJ-Gauss Core Theory

Cole-Hopf Transformation  $\rightarrow$  Linear Heat Equation



$\rightarrow$  Feynman-Kac Formula  $\rightarrow$  Picard Iteration.

## The Theoretical Roadmap of the Method

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1. Start from the **viscous** HJ PDE (Part A).
2. Apply a generalized **Cole-Hopf** transform  $\omega = e^{-cv} \rightarrow$  a **Linear heat equation**.
3. Solve the heat equation by its **Gaussian heat kernel**  $\rightarrow$  a **Feynman-Kac expectation**.
4. Recover value (log-sum-exp) and gradient from that expectation.
5. Because  $c$  depends on the unknown  $Dv$ , **iterate**: freeze, solve, update (Picard) = **Algorithm 1**.
6. Analyze exactness (quadratic case) vs the **quasi-linearization residual** (general case).

Each arrow is a theorem or proposition in the paper; we walk them one by one.

## The Transformation, and the Closed-Form/Approximate Split

For a general  $H$ , define the spatially-varying coefficient

$$c(t; x) = \frac{2}{\delta} \cdot \frac{H^\delta}{|Dv^\delta|^2}, \quad H^\delta := H(t; x, Dv^\delta),$$

and set  $\omega^\delta := \exp(-c v^\delta)$ .

- **Quadratic case**  $H = \frac{1}{2}|p|^2$ :  $c = 1/\delta$  constant,  $\omega^\delta$  solves the **homogeneous heat equation** with **zero residual** — a genuine Cole-Hopf identity.
- **General  $H$** : the transform induces a residual  $R = R_{\text{alg}} + R_{\text{der}}$ ;
  - $c$ 's choice eliminates the **algebraic** part  $R_{\text{alg}}$ , leaving the **derivative** part  $R_{\text{der}}$ .

No freezing of  $c$ 's *values* removes its *derivatives*. Hence for nonlinear  $H$  this is a **quasi-linearization**, not an identity — and the leftover  $R_{\text{der}}$  is the residual we will bound.

## Picard Quasi-Linearization: The Surrogate View

- At iteration  $k$ , **freeze**  $c^{(k)}$  at the current iterate and solve the linear heat equation **in closed form**.
- Equivalently: replace the true Hamiltonian by the **locally-matched quadratic surrogate**

$$\tilde{H}^{(k)}(p) = \frac{\delta}{2} c^{(k)} |p|^2,$$

solve the resulting viscous HJ equation in closed form, obtain  $v^{(k+1)}$ , update  $Dv^{(k+1)}$  and  $c^{(k+1)}$ .

- **The iteration's limit is the fixed point of the surrogate solve map** — equal to the viscous solution *only* when  $R_{\text{der}} \equiv 0$  (the quadratic case).

"Successive quadratic matching": each step is a locally closed-form quadratic-Hamiltonian solve, refined by re-estimating the co-state.

## Reduction to the Heat Equation (Proposition)

With  $c^{(k)}$  frozen,  $\omega^\delta$  solves the heat initial-value problem

$$\omega_t^\delta - \frac{\delta}{2} \Delta \omega^\delta = 0 \quad \text{in } \mathbb{R}^n \times (0, T], \quad \omega^\delta(0, x) = \exp(-c g(x)).$$

- When  $H = \frac{1}{2}|p|^2$ ,  $c = 1/\delta$ : **closed-form, no residual**.
- For general  $H$ : discards the derivative residual (Lemma, appendix) — the quasi-linearization.

A nonlinear, nonconvex, state-dependent HJ PDE has become **the heat equation** — the most classical linear parabolic PDE, with an explicit Gaussian solution.

## The Gaussian Heat Kernel & Feynman-Kac

The heat IVP has the unique bounded Green's-convolution solution

$$\omega^\delta(t, x) = \frac{1}{(\sqrt{2\pi\delta t})^n} \int_{\mathbb{R}^n} e^{-\frac{|x-y|^2}{2\delta t}} e^{-c g(y)} dy,$$

which, by **Feynman-Kac**, is a **Gaussian expectation**:

$$\omega^\delta(t, x) = \mathbb{E}_{y \sim \mathcal{N}(x, \delta t I_n)} [e^{-c g(y)}].$$

- The free-space kernel integrates to one over  $\mathbb{R}^n$  (and no proper subset) — every estimator is posed on all of  $\mathbb{R}^n$ .
- Sampling  $y \sim \mathcal{N}(x, \delta t I_n)$  and averaging  $e^{-c g(y)}$  is an **unbiased** estimator of  $\omega^\delta$ .

The solution is a **Gaussian roll-out average of the terminal cost**. This is the entire computational payload — no grid, no marching.

## Recovering the Value (Lemma)

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Invert the transform:

$$v^\delta(t, x) = -\frac{1}{c^{(k)}} \log \mathbb{E}_{y \sim \mathcal{N}(x, \delta t I_n)} [e^{-c^{(k)} g(y)}].$$

- A **log-sum-exp** of Gaussian-sampled terminal costs.
- Numerically evaluated in a **log-domain-stable** form (log-mean-exp) to avoid overflow/underflow.
- $M$  arbitrary evaluation states  $x_1, \dots, x_M$  — laid on a grid or scattered — each with its own fresh  $N$  samples.

The value at a query point is a scalar reduction over  $N$  samples. Embarrassingly parallel across both  $M$  states and  $N$  samples.

## Recovering the Gradient (Corollary)

The spatial gradient (co-state) has a closed Gaussian-expectation form:

$$Dv^\delta = \frac{1}{t \delta c^{(k)}} \left( x - \frac{\mathbb{E}_{y \sim \mathcal{N}(x, \delta t I_n)}[y e^{-c^{(k)} g(y)}]}{\mathbb{E}_{y \sim \mathcal{N}(x, \delta t I_n)}[e^{-c^{(k)} g(y)}]} \right).$$

- A **weighted-mean shift**: the gradient points from  $x$  toward the exponentially-weighted centroid of the samples.
- No finite differencing, no stored field — the co-state is a byproduct of the same samples.

This gradient is (up to constants) a **score function**  $\nabla \log \omega$ . Hold that: it is the entire bridge to diffusion in Part 7.

## Admissible Data & Unbiasedness (Recap as a Guarantee)

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- Datum  $\omega^\delta(0, \cdot) = e^{-cg}$  is continuous, strictly positive, bounded above by  $e^{-cg_{\min}}$ , decaying at infinity.
- $\Rightarrow$  the heat solution is the **unique bounded** one; the integral converges absolutely; the Gaussian expectation is **genuine and unbiased**.
- Only  $g \geq g_{\min}$  is required (bounded target); no upper bound on  $g$ .

The estimator drawing  $y \sim \mathcal{N}(x, \delta t I_n)$  over all of  $\mathbb{R}^n$  is **unbiased for the quantity inside the logarithm** — the foundation for the concentration bound of Part 4.

## C Sign and the Removable Zero of the Coefficient

- $c \sim C_H / (\delta |Dv^\delta|)$ : it **grows without bound** in flat regions and **vanishes and flips sign** where optimal dynamics run tangent to the level set (the barrier).
- The zero of  $c$  is a **removable** singularity of the estimator: as  $c \rightarrow 0$ ,

$$-\frac{1}{c} \log \mathbb{E}[e^{-cg}] \rightarrow \mathbb{E}[g],$$

i.e. the log-sum-exp gracefully degenerates to the **Gaussian mean** of the terminal data — the correct pure-diffusion limit ( $H \rightarrow 0$  reduces the PDE to the heat equation).

- In implementation: evaluate in a **log1p**-stable form and **clip**  $|c|$  to  $[c_{\min}, c_{\max}]$ .

The coefficient's pathologies are analytically understood and numerically tamed — clipping is itself a bounded, priced perturbation (next slide).

## Coefficient Regularization (Lemma)

Run Algorithm 1 with the **regularized coefficient**

$$c_\eta(t; x) = \frac{2}{\delta} \cdot \frac{H^\delta}{|Dv^\delta|^2 + \eta},$$

equivalent to running it unregularized for the **perturbed Hamiltonian**  $H_\eta = H |Dv^\delta|^2 / (|Dv^\delta|^2 + \eta)$ , which satisfies a **uniform** bound

$$\|H_\eta - H\|_\infty \leq \frac{C_H \sqrt{\eta}}{2},$$

so the induced value-function perturbation is at most  $TC_H \sqrt{\eta}/2$ .

Regularization cures the flat-interior blow-up at a **quantified, uniform** price — no hidden instability.

## Algorithm 1: Quasi-Linearization (Cole-Hopf)

$$\text{Fix } \varepsilon > 0, \quad v^{(0)}(t, x) = g(x), \quad c^{(0)} = \frac{2 H(t, x, Dg)}{\delta |Dg|^2}.$$

**for**  $k = 0, 1, 2, \dots$ :

(1) freeze  $c^{(k)}$  at the current iterate;

(2) solve  $\omega_t = \frac{\delta}{2} \Delta \omega$ ,  $\omega^{(k)}(0, x) = e^{-c^{(k)} g(x)}$ ;

$$\Rightarrow \omega^{(k+1)}(t, x) = \mathbb{E}_{y \sim \mathcal{N}(x, \delta t I)} [e^{-c^{(k)} g(y)}] \quad (\text{Monte-Carlo});$$

(3) recover  $v^{(k+1)} = -\frac{1}{c^{(k)}} \log \omega^{(k+1)}$ ;

(4) update  $Dv^{(k+1)}$ ,  $c^{(k+1)} = \frac{2 H(t, x, Dv^{(k+1)})}{\delta |Dv^{(k+1)}|^2}$  clip to  $[c_{\min}, c_{\max}]$ ;

(5) stop when  $\|v^{(k+1)} - v^{(k)}\| / \|v^{(k)}\| < \varepsilon$ .

- Steps (2)-(4) are all **Gaussian-expectation reductions** over  $N$  samples.
- Convergence typically in  $\leq 20$  iterations (empirically 12-15).

Every line is grid-free. The only state carried between iterations is the coefficient field  $c^{(k)}$  at the  $M$  evaluation states.

## Evaluation States vs Monte-Carlo Samples (Remark)

- $N$  = i.i.d. Gaussian draws  $y_1, \dots, y_N \sim \mathcal{N}(x, \delta t I_n)$  used **independently at each** evaluation state  $x$  to form the estimators.
- Resampled **fresh** at every evaluation state and every Picard iteration.
- $M$  = the evaluation states  $x_1, \dots, x_M$  where  $v$  is queried — carry **no randomness**, may be a uniform grid or scattered arbitrarily.
- Total randomness per iteration:  $M \times N$  **draws**.
- The zero level set is recovered by evaluating  $v^\delta$  at the  $M$  states and isocontouring at level zero (marching cubes) — a deterministic post-process consuming **no further samples**.

The **solve is grid-free**; the grids that appear in results are *evaluation windows* for visualization and error measurement, not sampling constraints.

## Memory: $O(N \cdot n)$ vs $O(M^n)$

|                | Grid level sets                          | HJ-Gauss                                     |
|----------------|------------------------------------------|----------------------------------------------|
| Representation | Stored field on $M^n$ grid               | $N$ Gaussian samples per query, $n$ -vectors |
| <b>Memory</b>  | $O(M^n)$                                 | $O(N \cdot n)$                               |
| Discretization | Cartesian grid + marching cubes to solve | None (grid-free solve)                       |
| Query states   | Tied to grid                             | Arbitrary / scattered                        |
| Parallelism    | Grid array ops                           | Over $M$ states <b>and</b> $N$ samples       |

- Example:  $N = 14,000$ ,  $n = 3 \rightarrow \approx 0.6$  MB/iteration;  $n = 45 \rightarrow \approx 7.2$  MB/iteration (vs  $10^{90}$  grid cells).

Memory now scales with **sample budget and dimension**, not with a grid raised to the dimension. This is the whole contribution in one line.

# Part 3

## Importance Sampling & Variance Control

Making the Gaussian estimator work in high dimensions.

## The Weight-Degeneracy Problem

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- The estimators are a **ratio of exponential-weight expectations**, e.g. the gradient's numerator/denominator both carry  $e^{-cg}$ .
- In high dimensions (or small  $\delta$ ),  $c g$  is large over the kernel support, so the denominator  $\mathbb{E}[e^{-cg}]$  is carried by a **rare event**.
- The weights collapse onto a few samples  $\rightarrow$  the ratio's **variance explodes**; the effective sample size crashes.

Naive Gaussian sampling is unbiased but can be catastrophically high-variance right where we need it (high  $n$ , small  $\delta$ ). We fix it with importance sampling.

## Tilted Proposal & Density-Ratio Reweighting

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- Replace  $y \sim \mathcal{N}(x, \sigma^2 I_n)$  ( $\sigma^2 = \delta t$ ) by a **shifted** proposal  $q_\theta = \mathcal{N}(x + \theta, \sigma^2 I_n)$ .
- Reweight by the **density ratio**

$$w_\theta(y) = \exp\left(\frac{|\theta|^2 - 2\langle y - x, \theta \rangle}{2\sigma^2}\right),$$

which leaves **every expectation unbiased**.

We move the samples to where the integrand has mass, then correct via the ratio. Bias is untouched; variance can collapse by orders of magnitude.

## The Zero-Variance Shift Is One Gradient Step

- The zero-variance proposal is proportional to the integrand  $\varphi_x(y) e^{-cg(y)}$  itself.
- Its first-order **Laplace / Gaussian** approximation gives the shift

$$\theta^* = -\sigma^2 c^{(k)} Dv^{(k)}(t, x).$$

- That is, samples are pushed **one preconditioned gradient step** along the descent direction of the running value iterate.

**The previous Picard iterate hands the sampler its drift for free.** Reachability and importance sampling become the *same* computation — the value gradient is the optimal sampling direction.

## Effective Sample Size as a Self-Diagnostic

- Monitor the **effective sample size** of the self-normalized weights  $\tilde{w}_i = w_\theta(y_i)e^{-cg(y_i)}$ :

$$\text{ESS} = \frac{(\sum_i \tilde{w}_i)^2}{\sum_i \tilde{w}_i^2} \in [1, N].$$

- $\text{ESS} \approx N$ : weights well-spread  $\rightarrow$  estimate certified.
- $\text{ESS} \rightarrow 1$ : **weight degeneracy**  $\rightarrow$  flag, increase  $N$ , raise  $\delta$ , or re-tilt.

ESS is a free, per-query health check: it tells you *when* to trust the estimate and *why* it failed when it does.

# Part 4

## Guarantees

Concentration · Contraction · Residual · Total Error · Certificate · Robustness.

## ▴ Four Error Sources, One Budget

The distance from the computed field  $\hat{v}^{K,N,\delta}$  to the true inviscid viscosity solution  $v$  decomposes into four independently-controlled sources:

| Source                                                        | Controlled by                   | Scaling                                |
|---------------------------------------------------------------|---------------------------------|----------------------------------------|
| <b>Iteration</b> (Picard truncation at $K$ )                  | More iterations                 | $C_1 \rho^K$                           |
| <b>Monte-Carlo</b> (finite $N$ )                              | More samples                    | $C_2 N^{-1/2} \sqrt{\log(1/\delta_p)}$ |
| <b>Quasi-linearization residual</b> $\mathcal{E}_{\text{ql}}$ | Surrogate fidelity (not effort) | Floor                                  |
| <b>Viscosity</b> (finite $\delta$ )                           | Smaller $\delta$                | $C_3 \sqrt{\delta}$                    |

Three of four shrink with computational effort; the **residual is a property of the surrogate** and sets the floor. We now bound each.

## ▴ Finite-Sample Concentration (Theorem)

Fix phase  $(t, x)$  and frozen  $c > 0$ ; assume  $g_{\min} \leq g \leq g_{\max}$  on the sampling support. With  $Z = e^{-cg(\zeta)}$ ,  $\mu = \mathbb{E}[Z]$ ,  $v_c = -\frac{1}{c} \log \mu$ ,  $\alpha = e^{-cg_{\max}}$ ,  $\beta = e^{-cg_{\min}}$ , and  $\hat{v}_{c,N} = -\frac{1}{c} \log \bar{Z}_N$ :

$$\mathbb{P}(|\hat{v}_{c,N} - v_c| \geq \varepsilon) \leq 2 \exp\left(-\frac{2N\mu^2(1 - e^{-c\varepsilon})^2}{(\beta - \alpha)^2}\right).$$

- Bounded weights  $\alpha \leq Z_i \leq \beta \Rightarrow$  Hoeffding.
- Rate is the standard  $O(N^{-1/2})$ , **independent of state dimension  $n$**  — the crux of the scalability claim.

Dimension enters the cost *per sample*, not the *number of samples* for a target accuracy. That is how we beat  $O(M^n)$ .

## Explicit Sample Size, and Its Worst Case (Corollary + Remark)

- Since  $\mu \geq \alpha$ , it suffices to take

$$N \geq \frac{(\beta - \alpha)^2}{2\alpha^2(1 - e^{-c\varepsilon})^2} \log \frac{2}{\alpha}$$

to guarantee error  $\geq \varepsilon$  with probability  $\leq \alpha$ .

- **Worst-case caveat:** with  $c = 1/\delta$ ,  $\beta/\alpha = e^{(g_{\max} - g_{\min})/\delta}$ , so this bound grows like  $e^{2(g_{\max} - g_{\min})/\delta}$  — **exponential in  $1/\delta$** , while viscosity error is only  $O(\sqrt{\delta})$ .

The accuracy gain from shrinking  $\delta$  and the sample cost pull **in opposite directions**. This Hoeffding bound is loose; the next slide tightens it, and experiments sit far below it.

## Tightening: Bernstein & Jensen

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- **Bernstein:** when  $\text{Var}(Z) \ll (\beta - \alpha)^2/4$ , a variance-aware tail bound gives strictly tighter concentration than Hoeffding.
- **Jensen:** the loose lower bound  $\mu \geq \alpha = e^{-c g_{\max}}$  can be replaced by  $\mu \geq e^{-c \mathbb{E}[g(\zeta)]}$ , substantially reducing the required  $N$ .
- **Importance sampling** (Part 3) attacks the same variance directly.

The pessimistic  $e^{1/\delta}$  is a worst-case artifact of Hoeffding + crude  $\mu$  bound. In practice  $N \in [14\text{k}, 20\text{k}]$  suffices at  $\delta \in [0.08, 0.1]$ .

## The Iteration Operator (Contraction Setup)

- Fix evaluation states  $x_1, \dots, x_M$  and time  $t$ ; let  $\mathcal{G}$  be a stable gradient-reconstruction operator,  $p_m(v) = (\mathcal{G}v)_m$ .
- Coefficient-update map:  $(\Gamma(v))_m = 2H(t, x_m, p_m(v)) / (\delta |p_m(v)|^2)$ .
- Frozen-coefficient heat-kernel map:

$$(\Phi(c))_m = -\frac{1}{c_m} \log \mathbb{E}_{\zeta \sim \mathcal{N}(x_m, \delta t I_n)} [e^{-c_m g(\zeta)}].$$

- Algorithm 1 is the iteration  $v^{(k+1)} = \Lambda(v^{(k)})$ ,  $\Lambda = \Phi \circ \Gamma$ .

The whole method is a fixed-point iteration of a composed map on  $\mathbb{R}^M$  with the sup-norm. Contraction  $\Rightarrow$  convergence.

## Assumptions for Contraction (Assumption A)

On a closed admissible set  $\mathcal{A}$  (sup-norm), assume:

1.  $|g| \leq G$ ;
2. non-degenerate gradient & bounded coefficient:  $m_0 \leq |p_m| \leq P_*$ ,  $c_{\min} \leq (\Gamma v)_m \leq c_{\max}$ ;
3. Hamiltonian Lipschitz in co-state:  $|H| \leq H_*$ ,  $|H(\cdot, p) - H(\cdot, q)| \leq L_H |p - q|$ ;
4. reconstruction Lipschitz:  $\|\mathcal{G}v - \mathcal{G}w\|_\infty \leq L_D \|v - w\|_\infty$ ;
5. invariance + contraction constant

$$q = \frac{2G}{c_{\min}} \cdot \frac{2L_D}{\delta} \left( \frac{L_H}{m_0^2} + \frac{2H_*P_*}{m_0^4} \right) < 1.$$

Non-degeneracy (item 2) holds on the **tube band**  $\{|v^\delta| \leq \eta\}$  where the gradient is bounded away from zero — where the barrier lives. Outside, use the regularized coefficient.

## Contraction Convergence (Theorem)

$\Lambda$  is a **contraction** on  $\mathcal{A} \Rightarrow$  unique fixed point  $v^\star$ , and for any  $v^{(0)} \in \mathcal{A}$ :

$$\|v^{(k+1)} - v^\star\|_\infty \leq q \|v^{(k)} - v^\star\|_\infty \leq q^{k+1} \|v^{(0)} - v^\star\|_\infty,$$

with the **a posteriori** estimate

$$\|v^{(k)} - v^\star\|_\infty \leq \frac{q}{1 - q} \|v^{(k)} - v^{(k-1)}\|_\infty.$$

- Iteration count for tolerance  $\varepsilon$ :  $K = O(\log(1/\varepsilon))$  — **geometric** convergence.

The residual you *watch* (successive-iterate change) bounds the error you *cannot* see (distance to  $v^\star$ ).  
That is the a posteriori estimate — a runtime stopping certificate.

## Scope of the Contraction (Remark)

- The theorem is about the **frozen-coefficient numerical map**, not the original nonlinear PDE.
- The fixed point  $v^*$  solves the viscous HJ equation with  $H$  replaced by its **quadratic surrogate at  $v^*$** .
- $v^* = v^\delta$  iff  $H = \frac{1}{2}|p|^2$  (then  $c \equiv 1/\delta$ ,  $R_{\text{der}} \equiv 0$ ).
- For general  $H$ , the two differ by the **quasi-linearization residual**  $\mathcal{E}_{\text{ql}} = \|v^* - v^\delta\|_\infty$ .

We do **not** claim unconditional global convergence to the true HJ solution. We claim geometric convergence to a surrogate fixed point, plus a **bound on the gap** — next.

## ▴ The Quasi-Linearization Residual (Theorem, via Duhamel)

With converged coefficient  $c^\star$ , coefficient-variation bound  $L_c$ , value bound  $G_v$ :

$$\mathcal{E}_{\text{ql}} = \|v^\star - v^\delta\|_\infty \leq \frac{T}{c_{\min}} e^{2c_{\max}G_v} (\bar{R}_{\text{alg}} + \bar{R}_{\text{der}}),$$

- $\bar{R}_{\text{alg}} \leq c_{\max}(L_H + 2H_*P_*/m_0^2) \|Dv^\delta - Dv^\star\|_\infty$  — **contracts as the surrogate gradient tracks the true one.**
- $\bar{R}_{\text{der}} \leq L_c G_v(1 + \frac{\delta}{2}) + \frac{\delta}{2} L_c^2 G_v^2 + \delta(1 + c_{\max}G_v)L_c P_*$  — governed by **coefficient variation**  $L_c$ .

**Proof idea:**  $\tilde{\omega} = e^{-c^\star v^\delta}$  solves the heat equation with a source  $\omega B$  (Lemma); Duhamel + heat-semigroup  $L^\infty$ -contraction bounds  $\|\tilde{\omega} - \omega^\star\|$ ; mean-value theorem for log converts back to  $v$ .

The bound **vanishes in the quadratic case** ( $L_c = 0, Dv^\star = Dv^\delta$ ) and is largest where  $H/|Dv|^2$  turns over sharply — the usable-part boundary.

## Reading the Residual

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- **Vanishes** when  $H = \frac{1}{2}|p|^2$ : recovers the closed-form Cole-Hopf.
- **Governed by**  $L_c$ , the variation rate of the converged coefficient — largest near the barrier where the Hamiltonian-to-gradient ratio shocks.
- **Algebraic part**  $\propto \|Dv^\delta - Dv^\star\|$ : the residual contracts as the Picard iteration refines the co-state.
- Certifies smallness where  $c^\star$  is slowly varying; offers **no reprieve** where it is not.

This is why experimental errors concentrate at the zero-level-set boundary — the theory *predicts* the location of the worst error.

## The Total Error Bound (Theorem) & Bias-Variance

Chaining all four sources by the triangle inequality:

$$\|\hat{v}^{K,N,\delta} - v\|_{\infty} \leq C_1 \rho^K + \frac{C_2}{\sqrt{N}} \sqrt{\log(1/\delta_p)} + \mathcal{E}_{\text{ql}} + C_3 \sqrt{\delta}.$$

- **Bias-variance tradeoff in  $\delta$ :** viscosity bias  $\downarrow$  as  $\delta \downarrow$ , but MC variance  $\uparrow$ .
- Balancing the two nontrivial  $\delta$ -dependent terms yields the optimal

$$\delta \sim N^{-1/3} \Rightarrow \text{overall rate } O(N^{-1/6}),$$

slower than plain MC  $O(N^{-1/2})$  but **dimension-robust and scalable**.

A single decomposition tells you how to spend budget: iterate to kill  $\rho^K$ , sample to kill  $N^{-1/2}$ , choose  $\delta \sim N^{-1/3}$ , and accept the residual floor.

## The Conservative Safety Certificate (Corollary)

Let  $E$  be the total error budget. Declare state  $x$ :

- **SAFE** only if  $\hat{v}^{K,N,\delta}(x) > E$ ;
- **UNSAFE** only if  $\hat{v}^{K,N,\delta}(x) < -E$ ;
- **UNDETERMINED** otherwise ( $|\hat{v}| \leq E$ ).

With probability  $\geq 1 - \delta_p$ : **no unsafe state is certified safe** and **no safe state is certified unsafe**; all classification error is confined to a declared band of width  $2E$  about the boundary.

- The certificate **errs on refusal, never on admission**.
- This is the accuracy quotation: **worst-case sign-correctness with an explicit abstention band**, not an  $L^2$  average that flatters the interior.

## Robustness to Model Uncertainty

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- The scheme is **stable under small perturbations** of the Hamiltonian and terminal cost (robustness theorems, appendix): perturb  $H$  by  $\Delta H$  in sup-norm  $\Rightarrow$  value perturbed by  $\leq T \|\Delta H\|_\infty$  (Duhamel/comparison).
- Consequence: perfectly known dynamics and terminal costs are **not required** — real-world models with bounded error still yield certified-with-margin sets.
- The coefficient-clipping and regularization perturbations are of this priced form.

The method degrades **gracefully and quantifiably** under model error — a prerequisite for deployment on real robots with imperfect models.

# Part 5

## Experiments (from the Paper)

Rockets, Dubins, a 45D game, and  $10^5$  birds.



## Experimental Setup and Statistics — 1/2

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- **Hardware:** single CPU (Intel i7-14700K, 20 cores, 31 GiB RAM, Ubuntu 22.04); JAX on CPU backend.  
— Consistent with the memory-frugality claim.
- **Replication:** Each experiment re-solved with **30 independent Monte-Carlo seeds**;
  - Evaluation points and the LevelSetPy reference held fixed — only sampler randomness varies.  
Tables report mean  $\pm$  1 s.d.

Figures show one representative seed for clarity; tables report full 30-seed statistics. Multiplicity-controlled reporting throughout.



## Experimental Setup and Statistics — 2/2

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- **Significance:** Three families of **Holm-Bonferroni-corrected** tests at  $\alpha = 0.05$ :
  - (a) paired Wilcoxon of 30-seed-averaged MC field vs grid reference;
  - (b) one-sided  $t$ -test that per-seed  $L_{\text{rel}}^2$  lies below the Crandall-Lions bound  $\sqrt{\delta}$ ;
  - (c) cross-condition Mann-Whitney (Rockets vs Dubins; speed regimes).

Figures show one representative seed for clarity; tables report full 30-seed statistics. Multiplicity-controlled reporting throughout.

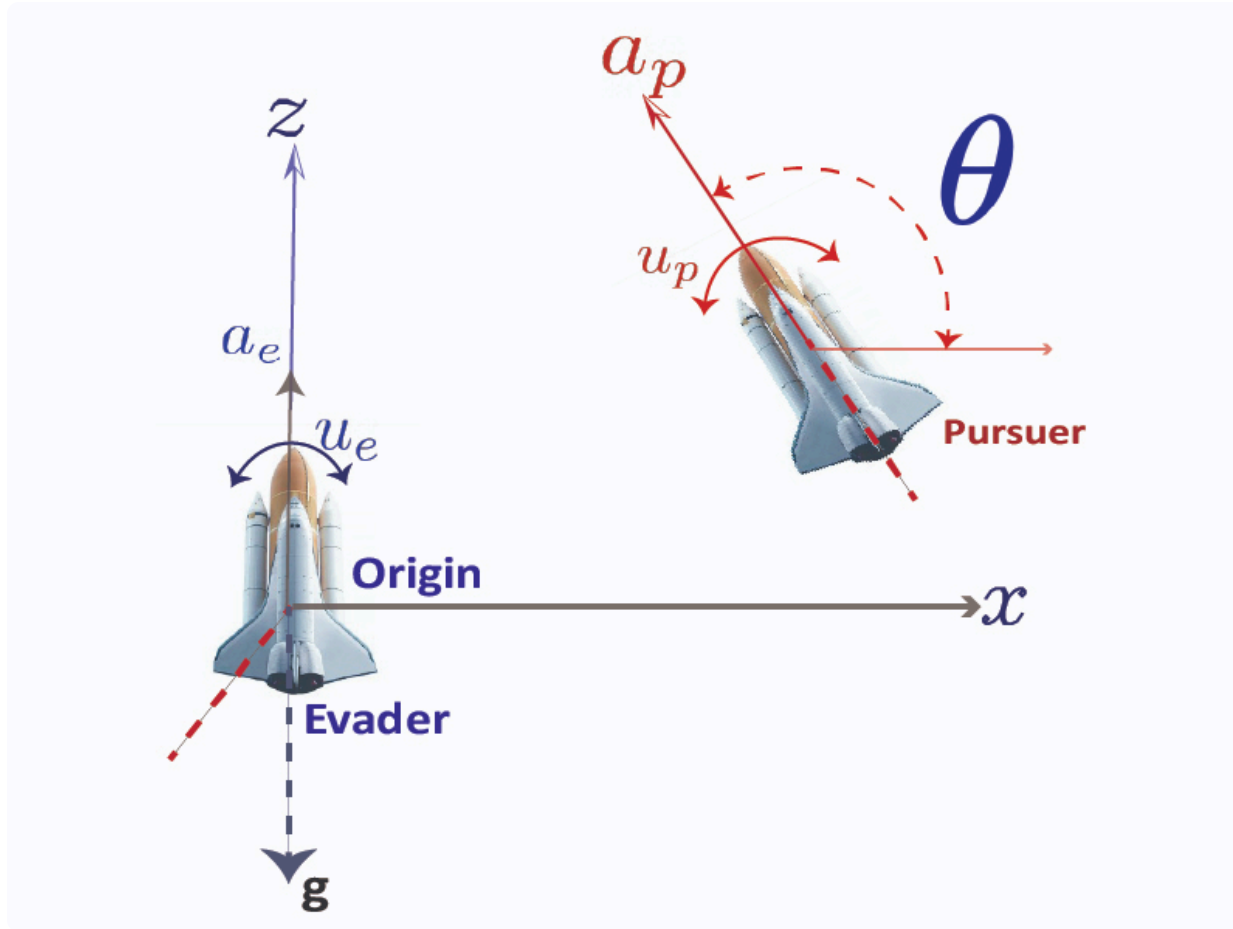
## Benchmark 1: Two-Vehicle Pursuit-Evasion Games

- **Rockets launch game**<sup>1</sup>: two identical rockets — pursuer  $P$ , evader  $E$  — on the  $x$ - $z$  plane, thrusts  $(u_p, u_e)$ , relative orientation  $\theta = u_p - u_e$ . Capture when  $\|PE\| < r$ .
- **Dubins two-car game** (Merz): relative  $(x_1, x_2, \theta)$ , symmetric turn-rate bound.
- Both are **3D relative-state** games; the BRT is a 3D tube, visualized as 2D  $(x, z)$  slices at fixed  $\theta$ .
- Target:  $\ell_2$ -ball of capture radius  $r = 1.5$ ; horizon  $(0, 1]$ ;  $\delta = 0.08$ ;  $N = 14,000$  (rockets) / 20,000 (Dubins) samples per iteration.

These are the canonical HJ-Isaacs benchmarks — and the Dubins game is the pairwise MAPF conflict of Part 6.

<sup>1</sup> Dreyfus 1966

## The Two-Rockets Problem (Schematic)



Relative geometry: evader  $E$  fixed at the origin (accel.  $a_e$ , control  $u_e$ , gravity  $g$ ); pursuer  $P$  at relative orientation  $\theta = u_p - u_e$  on the  $(x, z)$  plane. — HJ-Gauss (Molu et al., 2026).

## Two-Rockets: Dynamics and Value Function

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Relative state  $x = (x, z, \theta)$  — evader with respect to pursuer, with  $\theta = u_p - u_e$ :

$$\dot{x} = a_p \cos \theta + u_e x, \quad \dot{z} = a_p \sin \theta + a_e + u_e x - g, \quad \dot{\theta} = u_p - u_e.$$

Writing  $a \triangleq a_p = a_e$  and thrust bound  $\bar{u} = \max(|u_p|, |u_e|)$ , the Isaacs Hamiltonian is closed-form:

$$H(x, p) = -a p_1 \cos \theta - p_2 (g - a - a \sin \theta) - \bar{u} |p_1 x + p_3| + \bar{u} |p_2 x + p_3|.$$

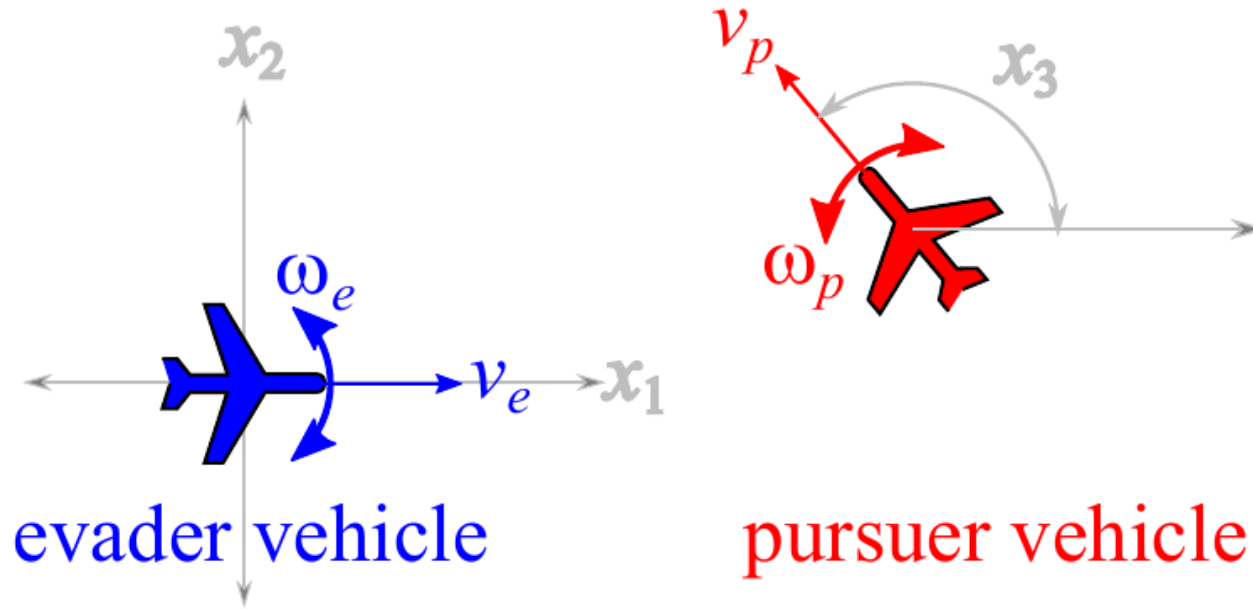
The capture tube is the viscosity solution of the variational HJ-Isaacs equation:

$$v_t(t, x) + \min\{0, H(t; x, Dv)\} = 0, \quad v(0, x) = \|(x, z)\| - r.$$

- The two  $|\cdot|$  terms **are** the optimal bang-bang thrusts, so no inner optimization is needed at run time.

## The Dubins Air3D Problem (Schematic)

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Air3D relative coordinates with the evader fixed at the origin: relative position  $(x_1, x_2)$  and relative heading  $\psi$ ; turn-rate-bounded pursuer vs evader. — LevelSetPy (Molu, ACM TOMS 2025 / IEEE

CDC 2024).

## Dubins Air3D: Dynamics and Value Function

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Relative state  $x = (x_1, x_2, x_3)$  — evader with respect to pursuer,  $x_3$  the relative heading:

$$\dot{x}_1 = -v_e + v_p \cos x_3 + \omega_e x_2, \quad \dot{x}_2 = -v_p \sin x_3 - \omega_e x_1, \quad \dot{x}_3 = -\omega_p - \omega_e.$$

With a shared turn-rate bound  $\omega$  (pursuer minimizing, evader maximizing), the Hamiltonian is again closed-form:

$$H(x, p) = p_1 (v_e - v_p \cos x_3) - p_2 v_p \sin x_3 - \omega |p_1 x_2 - p_2 x_1 - p_3| + \omega |p_3|.$$

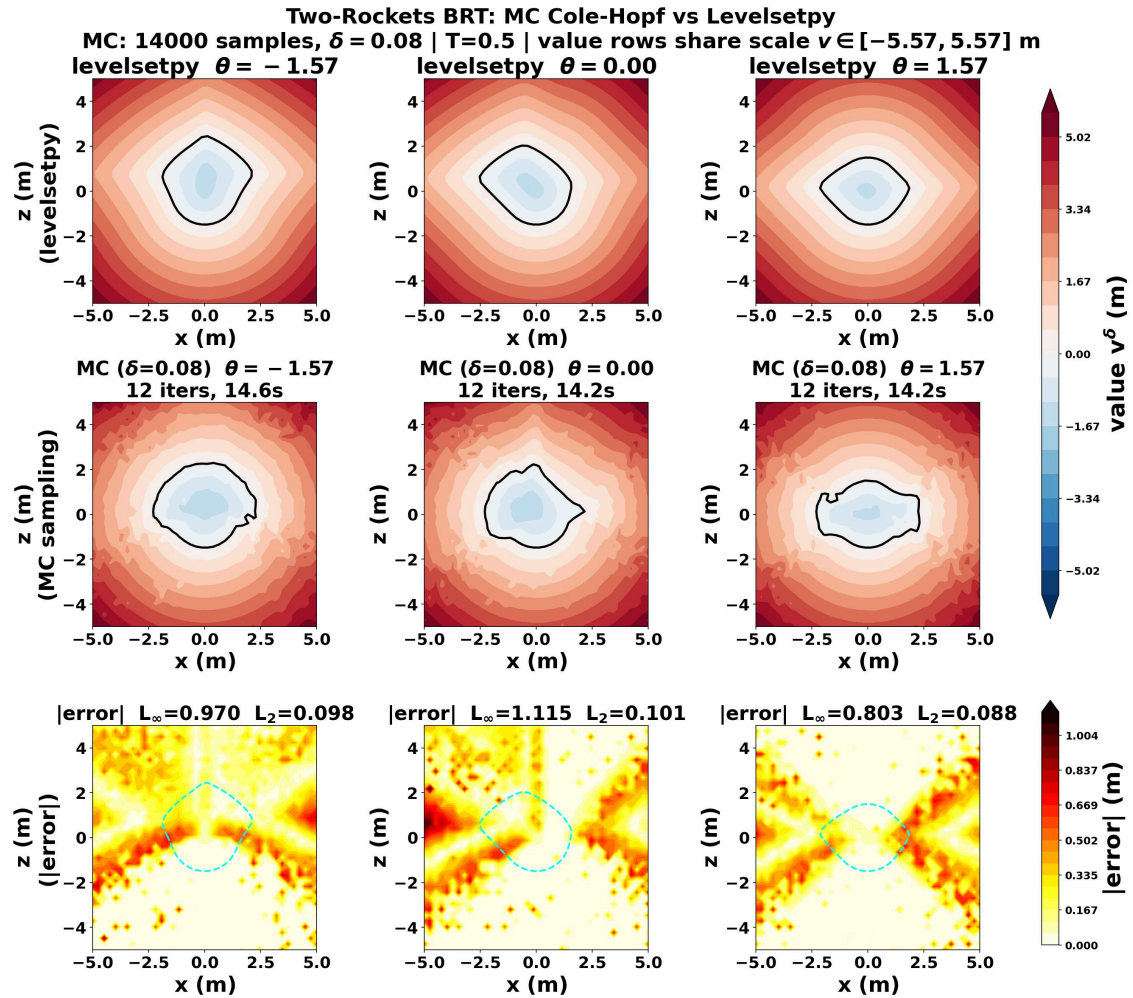
The conflict tube solves the same variational equation, with the capture cylinder as datum:

$$v_t(t, x) + \min\{0, H(t; x, Dv)\} = 0, \quad v(0, x) = \sqrt{x_1^2 + x_2^2} - r_c.$$

- The  $\min\{0, \cdot\}$  freeze is what makes the zero sublevel set a **tube** rather than a set: once captured, always captured.



# Rockets Pursuit-Evasion BRT



$(x, z)$  slices at  $\theta \in \{-90^\circ, 0^\circ, 90^\circ\}$ : LevelSetPy grid (top) · HJ-Gauss MC (middle) · pointwise error (bottom). — HJ-Gauss (Molu et al., 2026).



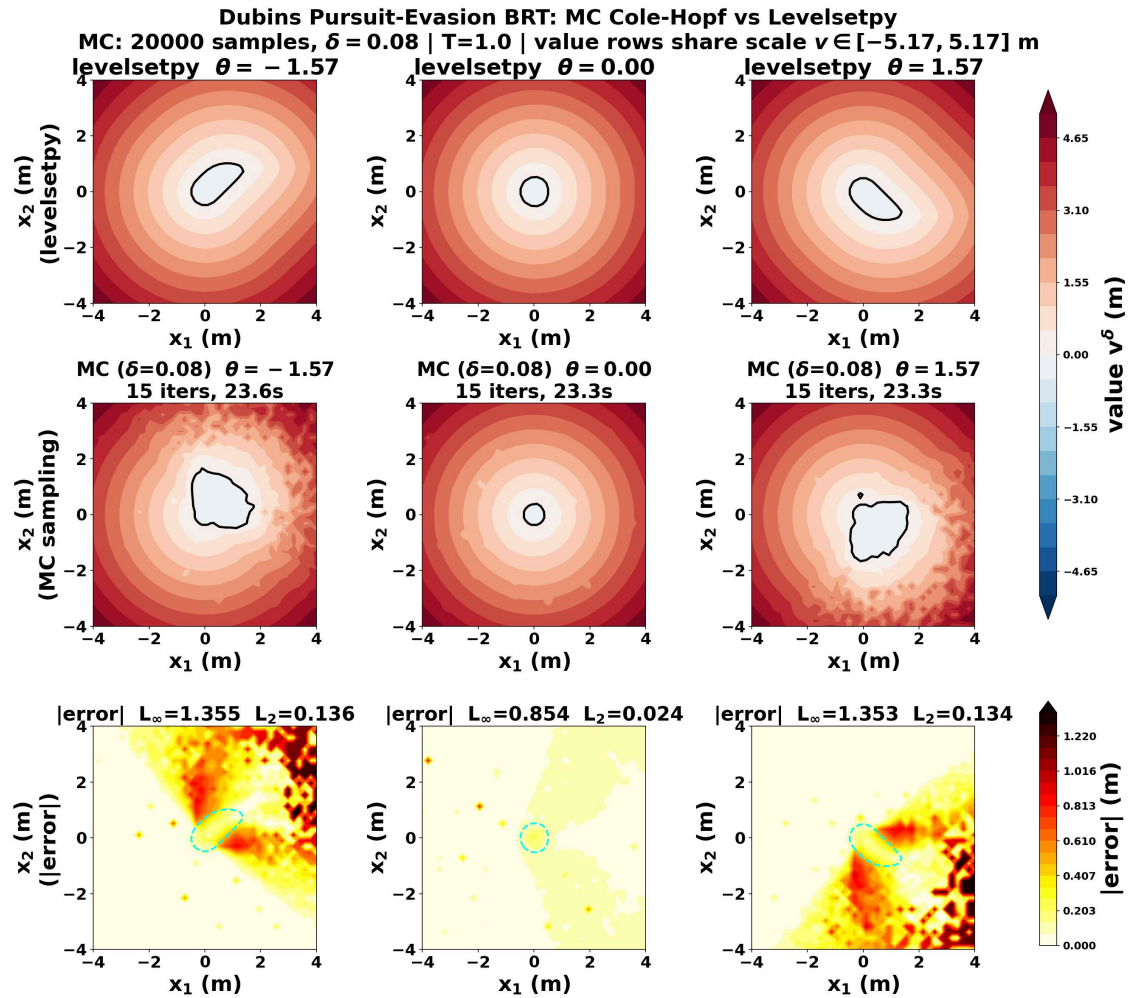
## Reading the Rockets Figure

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- **Layout:** the three columns are heading slices  $\theta \in \{-90^\circ, 0, 90^\circ\}$  of one 3D tube; the rows are **LevelSetPy grid** (top), **HJ-Gauss Monte-Carlo** (middle), and **pointwise error** (bottom).
- **What to look for:** the middle row recovers the top row's zero level set — the certified capture boundary — at every slice, with no grid ever stored.
- **Where the error lives:** the bottom row is near-zero across the interior and concentrates in a thin band **on the barrier**, exactly where the quasi-linearization residual bound says it should.
- **Physical read:** the  $\theta = -90^\circ$  and  $\theta = +90^\circ$  slices are **not** mirror images, because the gravity term  $(g - a - a \sin \theta)$  leaves a  $g - 2a$  drift at  $+90^\circ$  against  $g$  at  $-90^\circ$ .



# Dubins Two-Car BRT



LevelSetPy grid (top) · HJ-Gauss MC (middle) · error (bottom). — HJ-Gauss (Molu et al., 2026); LevelSetPy (Molu, TOMS 2025 / CDC 2024).



## Reading the Dubins Figure

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- **Layout:** the same three-row comparison — grid, Monte-Carlo, error — for the Air3D game at the same three relative headings.
- **The signature shape:** the classic Merz notch appears where the turn-radius bound denies escape; a swept-disc test cannot produce that shape at all.
- **Agreement:**  $L_{\text{rel}}^2$  runs 0.024 at  $\theta = 0$  (smooth interior) to  $\approx 0.13$  at  $\theta = \pm\pi/2$ , while  $L^\infty \approx 0.7$ -1.4 is set by the boundary where  $|Dv^\delta|$  peaks.
- **Why it matters here:** this tube **is** the pairwise conflict predicate the MAPF shield queries, so the collision reduction rests on this agreement.

## Benchmark 1: Quantitative Results

| System  | $\theta$ (rad) | $L^\infty$        | $L^2_{\text{rel}}$ | MC time (s) | Iters |
|---------|----------------|-------------------|--------------------|-------------|-------|
| Rockets | $-\pi/2$       | $0.855 \pm 0.047$ | $0.098 \pm 0.002$  | 13.5        | 12    |
| Rockets | 0              | $1.034 \pm 0.067$ | $0.101 \pm 0.001$  | 13.5        | 12    |
| Rockets | $\pi/2$        | $0.895 \pm 0.109$ | $0.090 \pm 0.002$  | 13.6        | 12    |
| Dubins  | $-\pi/2$       | $1.352 \pm 0.002$ | $0.131 \pm 0.003$  | 23.6        | 15    |
| Dubins  | 0              | $0.701 \pm 0.144$ | $0.024 \pm 0.001$  | 23.7        | 15    |
| Dubins  | $\pi/2$        | $1.352 \pm 0.002$ | $0.132 \pm 0.003$  | 23.7        | 15    |

- Reference: LevelSetPy  $45^3$  grid interpolated to a  $40 \times 40$  evaluation grid.
- Holm-Bonferroni one-sided  $t$ -test rejects  $L^2_{\text{rel}} \geq \sqrt{\delta} = 0.283$  at  $p_{\text{holm}} < 10^{-50}$  in **every** condition.

Sampling-plus-iteration error is statistically well inside the viscosity budget — not merely numerically.

## Interpreting the Errors: Different Currencies

- The **Crandall-Lions**  $O(\sqrt{\delta}) \approx 0.283$  bounds the *inviscid-vs-viscous* sup-distance; the table measures *MC-vs-grid* in  $L^2/L^\infty$  — **different currencies**, compared with care.
- **Averaging does not equal agreement**: a paired Wilcoxon of the 30-seed-averaged MC field vs the grid rejects equality at  $p_{\text{holm}} < 10^{-8}$  everywhere — a systematic **quasi-linearization residual** remains (Theorem, Part 4).
- $\theta = 0$  Dubins reaches  $L_{\text{rel}}^2 = 0.024$  (smooth interior);  $\theta = \pm\pi/2$  and rockets sit  $\sim 0.09$ - $0.13$  (share of grid near the coefficient-turnover boundary).
- $L^\infty \sim 0.7$ - $1.4$  driven by the zero-level-set boundary where  $|Dv^\delta|$  is maximal — where the residual  $L_c$  term predicts.

What we report is the **Corollary (conservative certificate)**: sign-correctness outside an abstention band;  $L^2$  flatters the interior,  $L^\infty$  is dominated by the declared-undetermined band.



## A Physical Check: The Gravity Asymmetry

- Rockets are asymmetric between  $\theta = -90^\circ$  and  $\theta = +90^\circ$ : the gravity term ( $g - a - a \sin \theta$ ) gives a  $g - 2a$  drift at  $+90^\circ$  but  $g$  at  $-90^\circ$ .
- Holm-Bonferroni test over 30 seeds detects this asymmetry for **Rockets** ( $p_{\text{holm}} < 10^{-9}$ ) but **not** for the gravity-free **Dubins** vehicle ( $p_{\text{holm}} = 0.53$ ).

The method resolves a **genuine physical effect**, and the statistics correctly find it present where physics says it should be and absent where it should not. Validation, not just fitting.



## Benchmark 2: A 45-Dimensional Pursuit-Evasion Game

- **15 rockets** (14 pursuers, 1 evader), state  $x \in \mathbb{R}^{45}$ : each agent  $(x_i, y_i, \theta_i)$ , control  $u_i \in [-1, 1]$ .
- Dynamics  $\dot{x}_i = a_i \cos \theta_i$ ,  $\dot{y}_i = a_i \sin \theta_i$ ,  $\dot{\theta}_i = u_i$ ; target  $\phi(x) = \min_{i \leq 14} \|x_i^{\text{pos}} - x_{15}^{\text{pos}}\| - r_{\text{capture}}$ .
- Three regimes: **evader faster** (2.0 vs 1.0), **balanced** (1.0/1.0), **pursuers faster** (1.0 vs 2.0).
- A 101-point grid would need  $101^{45} \approx 10^{90}$  cells; **HJ-Gauss operates at  $\approx 7.2$  MB per iteration**.

This is the flagship: a reachability computation **grid solvers cannot even represent**, done on a single CPU in seconds.



## Benchmark 2: 45D Results

| Case            | $a_{\text{evader}}$ | $a_{\text{pursuers}}$ | Iters | Residual $\varepsilon(k)$ | Wall-clock (s) |
|-----------------|---------------------|-----------------------|-------|---------------------------|----------------|
| Evader faster   | 2.0                 | 1.0                   | 15    | $0.0002 \pm 0.0005$       | 12.7           |
| Equal speed     | 1.0                 | 1.0                   | 15    | $0.0003 \pm 0.0005$       | 12.6           |
| Pursuers faster | 1.0                 | 2.0                   | 15    | $0.0006 \pm 0.0013$       | 12.6           |

- Memory constant at **7.2 MB/iteration**; residuals below  $10^{-2}$  (indeed  $\sim 10^{-4}$ ) → **near-zero Picard residual floor**.
- Min-max Hamiltonian evaluates in **closed form** (box-constrained control  $\Rightarrow$  sign structure), so each iteration is  $O(N \cdot n)$  arithmetic and memory.
- Pairwise Holm-Bonferroni Mann-Whitney: **no significant residual difference** between speed regimes ( $p_{\text{holm}} \geq 0.93$ ) → floor set by  $(N, \delta)$ , not game parameters.

At  $n = 45$  there is **no grid reference**; we report *measurable* quantities — iteration stability, memory, wall-clock — i.e. **scalability, not certified accuracy**.

## Why the 45D Residual Floor Is *Lower* Than 3D

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- 3D benchmarks floor at 0.02-0.06; the 45D game floors at 0.0002-0.0006 — **two orders of magnitude lower**.
- Reason: the residual floor tracks **proximity to the coefficient-turnover region** (the shock-prone barrier), not dimension per se.
- 45D evaluation states are drawn uniformly over  $[-100, 100]$  per coordinate — with **small probability of landing near the boundary** — so the average residual is small.

A subtle but important message: **error concentrates geometrically (at the barrier), not dimensionally**. High dimension is not inherently high error.



## Cost Accounting: Time vs Memory

- 3D isosurface over 15,625 points: **129.4 s** (rockets, single CPU) vs **1.3 s** for the dense  $45^3 = 91,125$ -point LevelSetPy grid.
- Per-iteration memory:  $\approx 0.6$  MB (MC,  $N = 14\text{k}$ ,  $n = 3$ ) vs  $\approx 1.5$  MB (grid value+gradient).
- **In low dimensions ( $n \leq 4$ ) grids win** — smaller constant, no sampling variance.
- The Monte-Carlo overhead **becomes favorable where grid storage becomes prohibitive** ( $n \geq 5$ ).

The two paradigms are **complementary, not competing**. Use grids where they fit in memory; use HJ-Gauss where they cannot.



## Benchmark 3: Safety at Population Scale ( $10^5$ Starlings)

- Certify **100,000** European starlings (*Sturnus vulgaris*) as 4D aerial Dubins vehicles under predator attack.
- **Structure (explicit)**: the murmuration is partitioned into **flocks**; each flock has its **own** value function; the population safe set is the aggregation of per-flock zero sublevel sets — **many coupled low-D games, not one joint high-D PDE**.
- Because the solve is grid-free, the value-function solve is **independent of bird count**; the population enters only through **parallel per-bird certification** — the operational content of  $O(N \cdot n)$  at  $10^5$  agents.

Two curses dodged at once: **dimensionality** (grid-free solve) and **agent cardinality** (per-flock decoupling + parallel evaluation).

## The Repertoire Is Field-Documented

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- Real starling responses to predation: **cordon**, **tube**, **funnel**, and **split**.
- Each one appears in our certificates as a **topological signature** of the safe set, and none of them is a hand-coded rule.
- The certificate reproduces a documented repertoire from **dynamics and the game alone** — which is the reason to trust it on a floor, where the repertoire is congestion, yielding, and deadlock instead.

Fragmentation event,  $n_c : 1 \rightarrow 2$ . Field murmuration imagery as reproduced in HJ-Gauss (Molu et al., 2026).



## 🐦 Behaviour 1: Dense Ascent From a Roost

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- Thousands of birds leave a roost at once, and the flock's **local density spikes** while every bird is still turn-rate and climb-rate limited.
- **The biological question:** can a bird in the interior still retain an escape maneuver when its near-neighbour spacing collapses?
- **What we certify:** whether the interior states stay outside the capture set for the whole window — the dense-packing analogue of a congested aisle.



Series photo credits: Reuters/Amir Cohen · AP Photo/Oded Balilty · Menahem Kahana/AFP/Getty Images · Courtesy of The

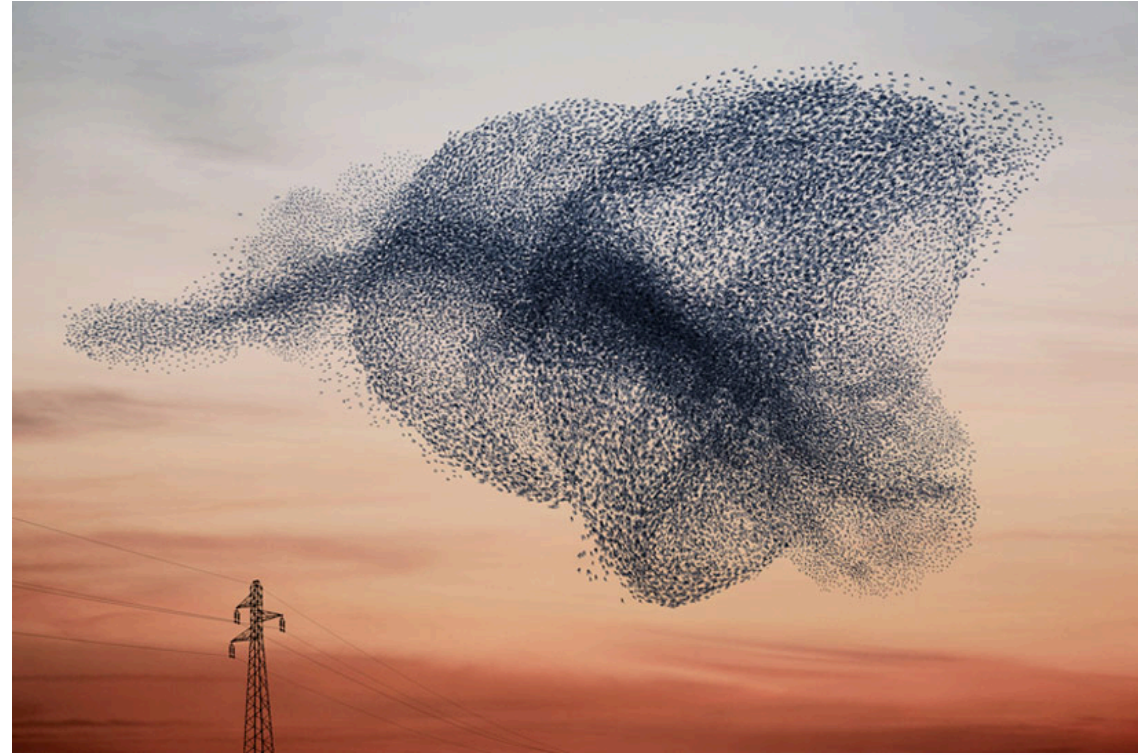
Gathering Site — reproduced in LevelSetPy (Molu, ACM TOMS 2025) and HJ-Gauss (Molu et al., 2026).



## Behaviour 2: The Concentric Cordon

- Under sustained pressure the flock arranges itself into a **concentric, tube-like shell** around an interior core.
- **The biological question:** is the core genuinely protected, or does it merely look protected from outside?
- **What we certify:** an **annular** safe set, first Betti number  $\beta_1 = 1$  — a hole the predator cannot certify entry into. This is the cordon marker.

Series photo credits as on the previous slide.





## Behaviour 3: Funnel and Split

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- A pressed flock **necks into a funnel** and can divide, leaving a subgroup separated from the main body.
- **The biological question:** which birds end up isolated, and when does isolation become unavoidable?
- **What we certify:** a rise in the connected-component count  $n_{\text{comp}}$  — fragmentation, the marker that says a subgroup has been cut off rather than merely crowded.



Series photo credits as on the first slide of this series.



## The Bird Model: 4D Aerial Dubins

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Each bird carries state  $(x_1, x_2, x_3, \theta) \in \mathbb{R}^2 \times \mathbb{R} \times \mathbb{S}^1$  — planar position, altitude, heading:

$$\dot{x}_1 = v \cos \theta, \quad \dot{x}_2 = v \sin \theta, \quad \dot{x}_3 = u_z, \quad \dot{\theta} = \langle \omega \rangle_r.$$

- The **coupling lives in the heading**: each bird turns at the neighbour-averaged rate

$$\langle \omega \rangle_r = \frac{1}{1 + n_i} \left( \omega_i + \sum_{j \in \mathcal{N}_i} \omega_j \right), \quad n_i = |\mathcal{N}_i|.$$

- Alignment is therefore a property of the **dynamics**, not a penalty added to a cost.
- Bounded controls:  $|u_z| \leq \gamma_{\max}$  for climb rate,  $|\omega| \leq \bar{\omega}$  for turn rate.

## The Attacked-Flock Game

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Relative state of an attacked bird with respect to its predator:

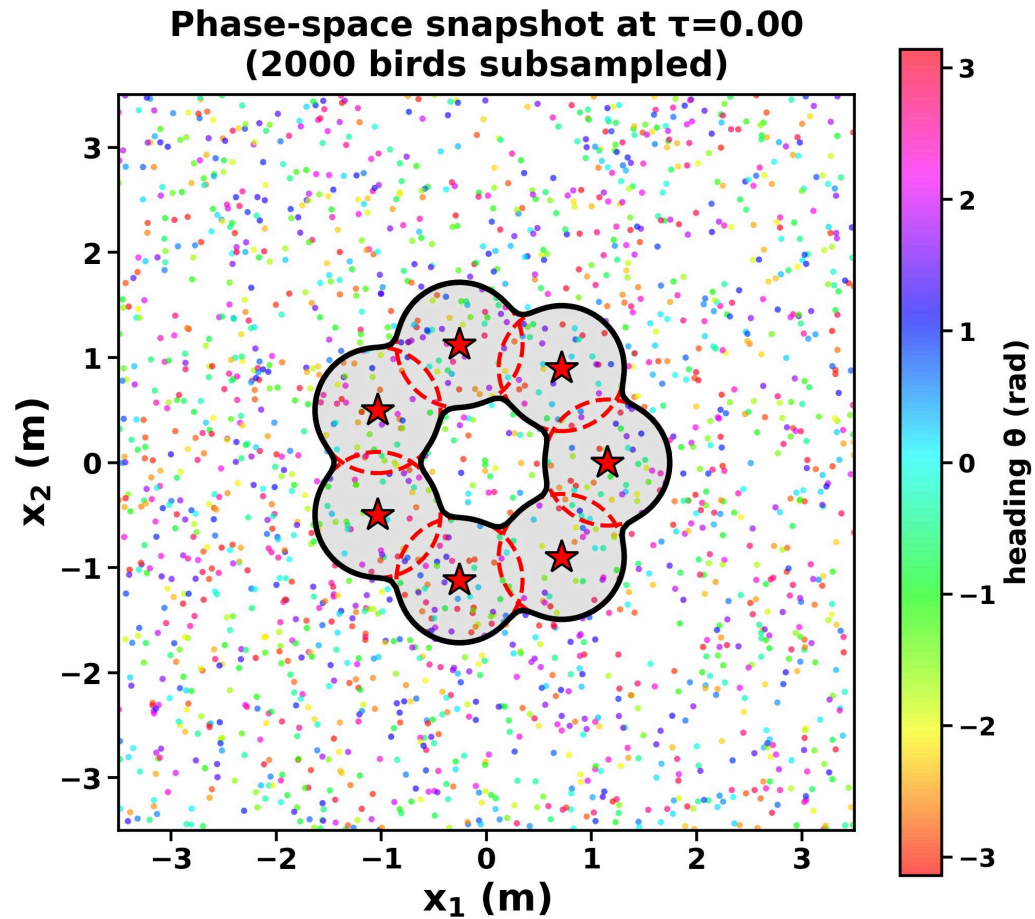
$$\begin{aligned}\dot{x}_1 &= -v_p + v_e \cos \theta + \langle \omega_e \rangle_r x_2, & \dot{x}_2 &= v_p \sin \theta - \langle \omega_e \rangle_r x_1, \\ \dot{x}_3 &= u_z^e - u_z^p, & \dot{\theta} &= \omega_p - \langle \omega_e \rangle_r.\end{aligned}$$

Datum is the capture cylinder  $g(x) = \sqrt{x_1^2 + x_2^2} - r_c$ , and each flock solves its **own** variational equation:

$$v_t + \min\{0, H_{\text{att}}(x, Dv)\} = 0, \quad H_{\text{att}} = \min_{u_p} \max_{u_e} Dv^\top f(x, u_p, u_e).$$

- Run shown here: **99,996 starlings** in six flocks, **7 predators** on a ring of radius 1.15,  $r_c = 0.6$ ,  $\delta = 0.18$ ,  $N = 1600$  samples,  $\leq 7$  Picard iterations,  $128 \times 128$  window, 18 backward-time steps.
- Whole study, value solves plus certifying every bird: **429 s on eight CPU cores**, no GPU. **87.7%** of birds end certified outside the capture set.

## The Certified Population Snapshot



2,000 of 99,996 birds (colored by heading) with 7 flock centers (stars) and their capture discs (dashed). Black contour is the certified safe-set boundary at  $\tau = 0$ : an **annular cordon** around a protected core. — HJ-Gauss (Molu et al., 2026).



## Reading the Population Snapshot

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- **Coloured dots:** 2,000 of the 99,996 birds, subsampled for legibility and coloured by heading  $\theta$ . The spread of colour is the point — the population is a **heading distribution**, not a rigid formation.
- **Stars and dashed circles:** the seven predators and their capture radii  $r_c = 0.6$ .
- **Black curve:** the  $v = 0$  boundary at  $\tau = 0$ . The shaded zero-sublevel set  $\{v \leq 0\}$  is the flock's **certified safe set**.
- **The hole is the message:** an annular safe set ( $\beta_1 = 1$ ) is a **defensive cordon** — a protected core the predators cannot certify entry into.



## What the Snapshot Certifies

- Each bird's status is one **membership query** against the cached value field: sign of  $v$  at its own 4D state.
- So the 99,996 birds cost 99,996 **independent**  $O(1)$  lookups, trivially parallel — while the solve itself never represented  $10^5$  agents.
- That is the operational meaning of  $O(N \cdot n)$ : the certificate's cost scales with the **sample budget and state dimension**, not with fleet size.

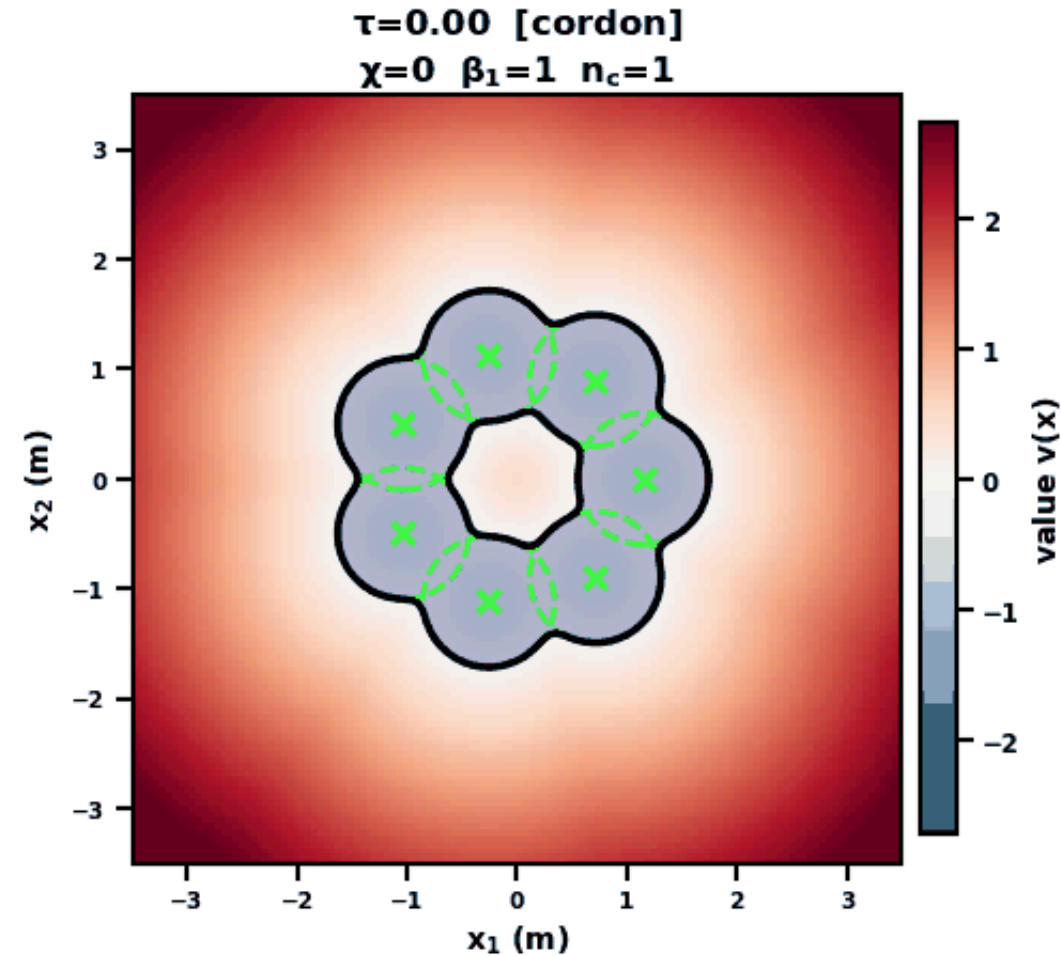
The warehouse analogue: certify a thousand drive units against a per-zone value field, with no joint state space anywhere in the loop.



## Watch the Safe Set Change Shape

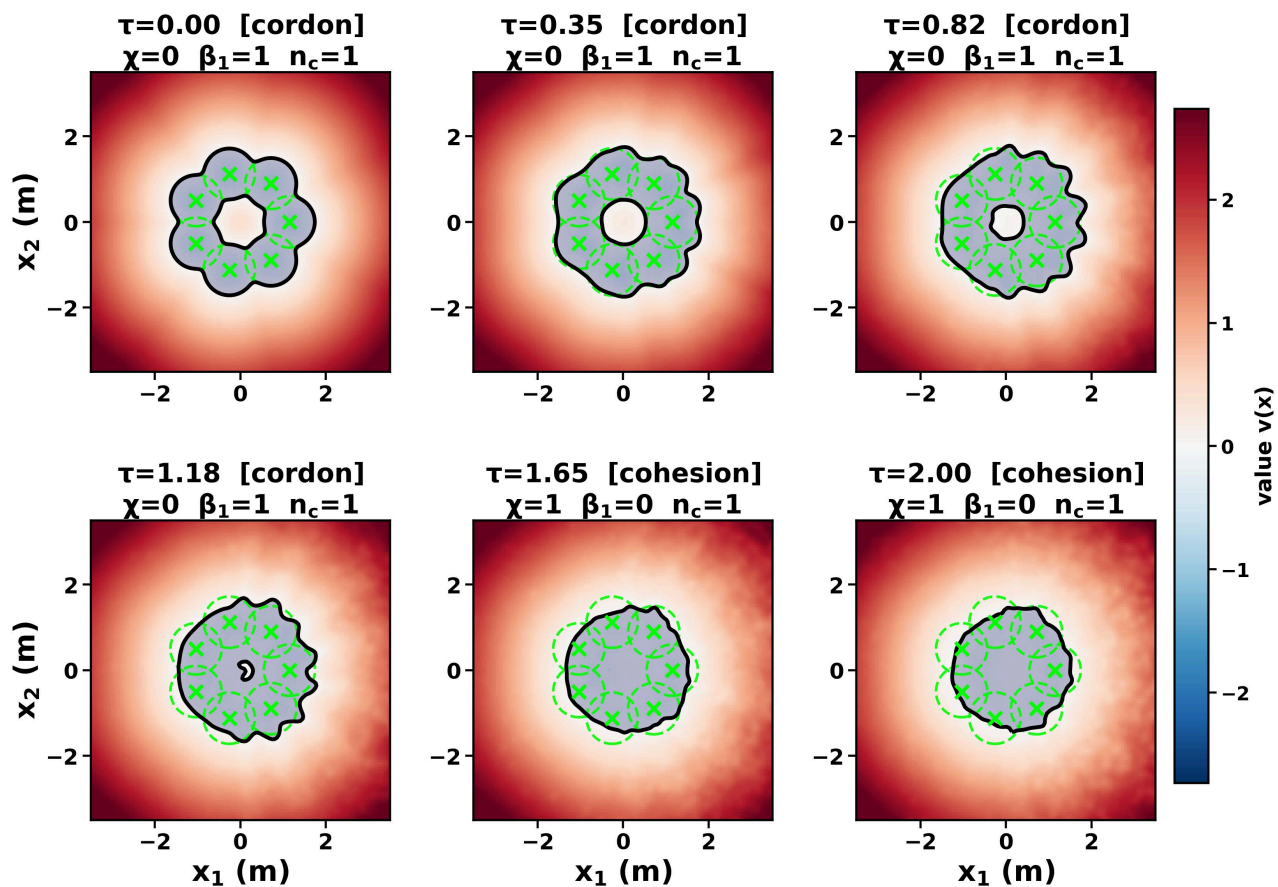
- Backward time  $\tau$  runs outward from the capture set: the certified safe set of the attacked flock **grows and changes topology**.
- Green crosses and dashed circles are the seven attacking predators with their capture discs.
- Blue interior = certified safe; the black curve is the  $v = 0$  boundary.

A safety certificate for  $10^5$  agents, as one evolving surface. (Same solve as the filmstrip, sampled at 24 backward-time steps.)



## Cordon, Then Collapse

### BRT zero-level-set evolution (backward time $\tau$ )



Same solve, six instants. A hole persists in the safe set ( $\chi = 0, \beta_1 = 1$ , cordon) through  $\tau = 1.18$ ; by  $\tau = 1.29$  it has closed ( $\chi = 1, \beta_1 = 0$ , cohesion). — HJ-Gauss (Molu et al., 2026).

## Topology as a Safety Instrument

The certificates recover the field-documented collective repertoire as **topological events** of the reachable set, compressed to three integers per time step  $(\chi, \beta_1, n_{\text{comp}})$ :

| Behavior                                        | Topological signature                              | Marker                       |
|-------------------------------------------------|----------------------------------------------------|------------------------------|
| <b>Vacuole nucleation</b> (predator penetrates) | Drop in Euler characteristic $\chi$                | Threshold crossing in $\chi$ |
| <b>Defensive cordon</b> (protected core)        | Annular safe set, first Betti number $\beta_1 = 1$ | $\beta_1$ transition         |
| <b>Flock fragmentation</b>                      | Rising connected-component count $n_{\text{comp}}$ | Component-count jump         |

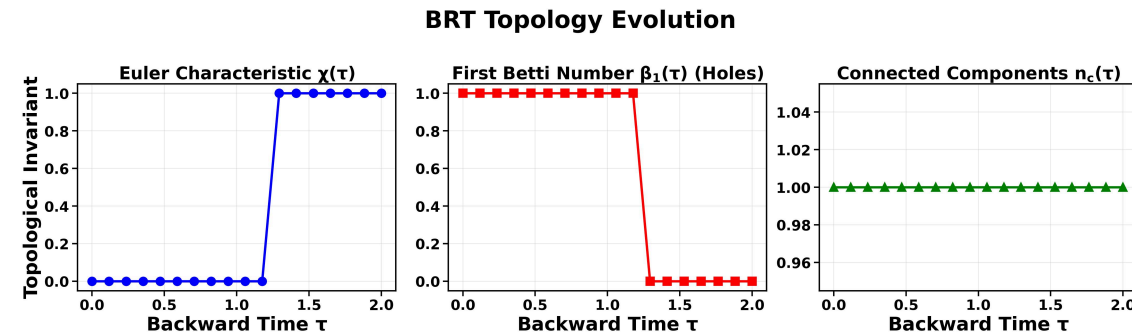
The triple  $(\chi, \beta_1, n_{\text{comp}})$  tells an operator not merely **that** safety is being lost but **how** — nucleation vs cordon vs split. Safety posture as a 3-integer summary.



## The Three Integers, Over Time

- $\chi$  steps  $0 \rightarrow 1$  while  $\beta_1$  drops  $1 \rightarrow 0$  at the **same**  $\tau = 1.29$ : one event, two independent witnesses.
- $n_c$  holds at 1 throughout, so this flock never fragments.
- **Vacuole nucleation** is the reverse event: a predator penetration attaches a 1-handle and drops  $\chi$  by one.

Three integers per tick say whether safety is being lost, and **how**.



# Part 6

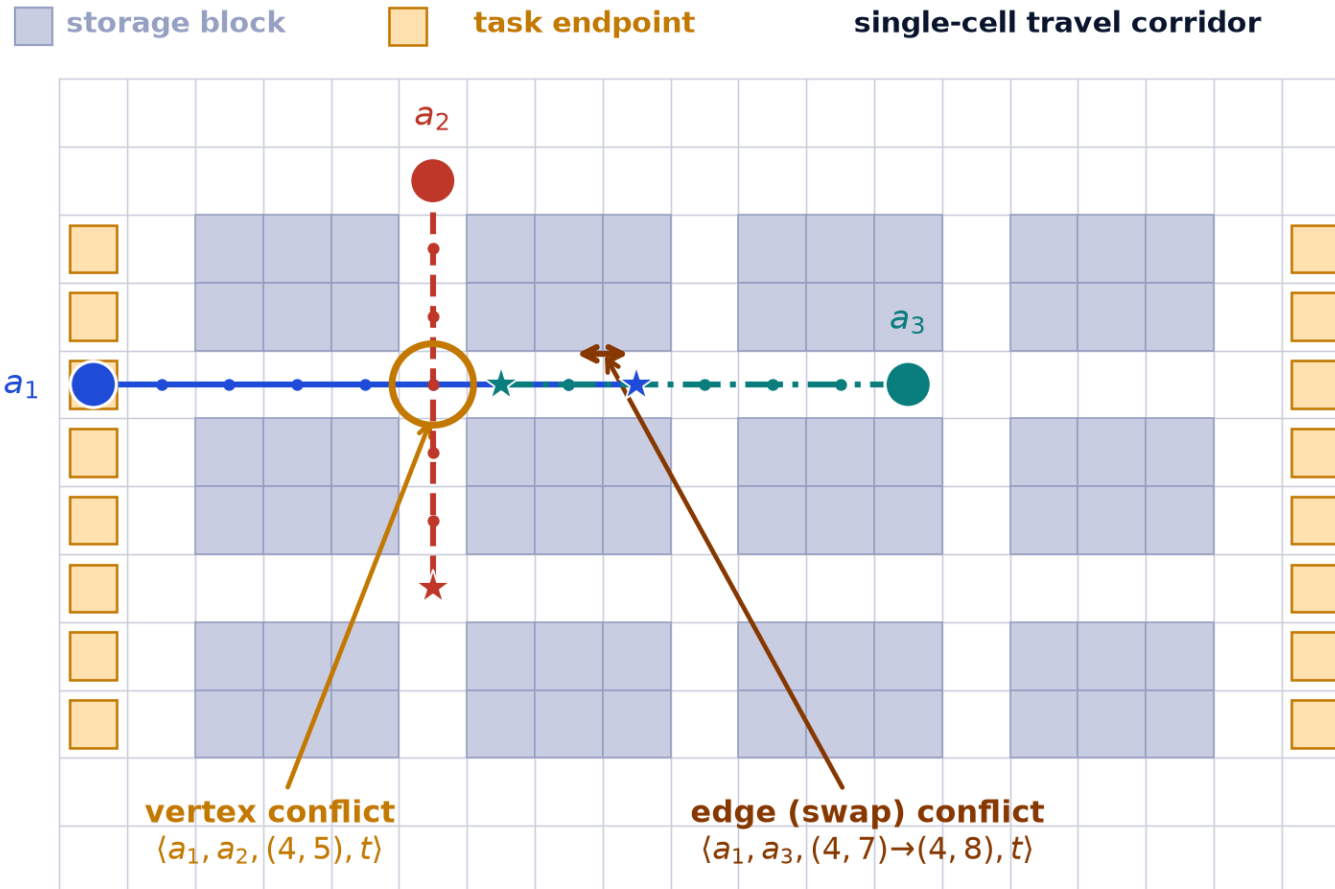
## Application: Multi-Agent Path Finding

The reason this talk is for the autonomous mobility team.



# The Floor, as MAPF Sees It

**Discrete MAPF: the grid-partitioned floor that CBS / ECBS / RHCR search over**



Storage blocks separated by single-cell travel corridors, task endpoints on the border columns — the RHCR fulfillment-warehouse map family (Li et al., AAAI 2021). CBS/ECBS resolve the two classical conflict types marked above.



## What Counts as a Conflict

---

- **Vertex conflict**  $\langle a_i, a_j, x, t \rangle$ : two agents occupy the same cell at the same timestep.
- **Edge (swap) conflict**  $\langle a_i, a_j, x \rightarrow y, t \rangle$ : two agents traverse the same edge in opposition — the head-on case in a one-cell corridor.
- **CBS** branches on a conflict, adds a constraint to one agent, and re-plans that agent's shortest path; **ECBS** bounds suboptimality; **RHCR** applies this inside a rolling window.
- Every one of these predicates is a statement about **cells and timesteps**.

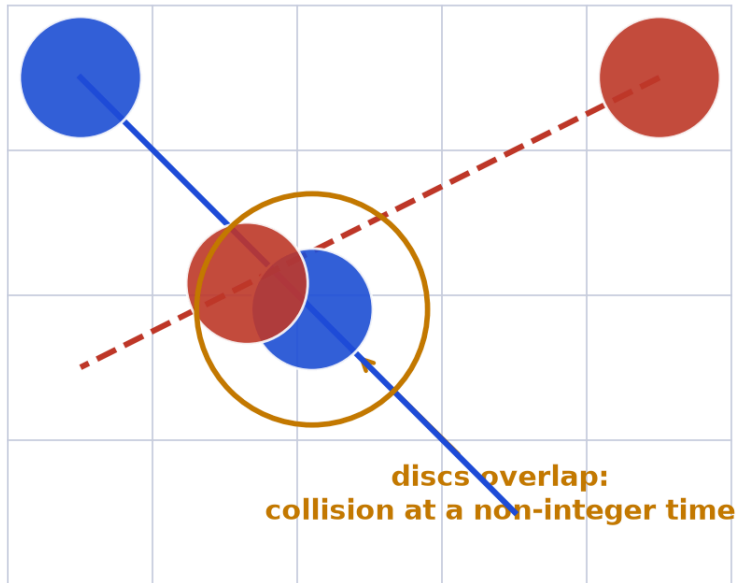
Nothing in this vocabulary mentions turn radius, actuation lag, or localization error.



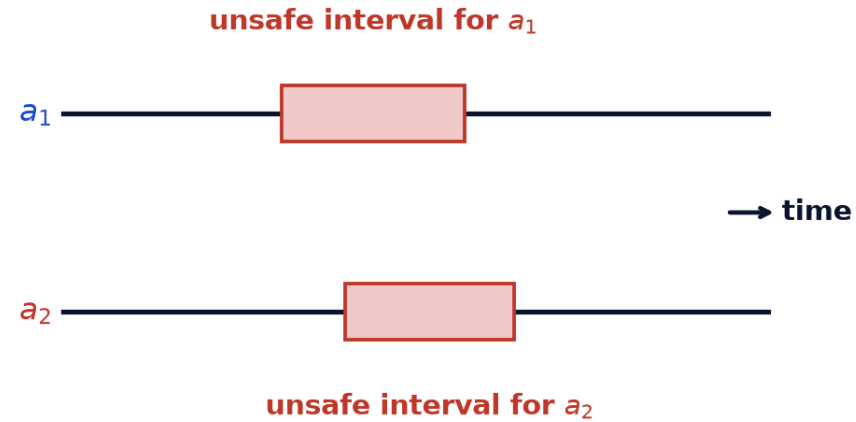
# CCBS Fixes Time, Not Dynamics

## CCBS: geometric agents in continuous time

straight-line moves, arbitrary durations,  $2^k$ -neighbourhood



## Conflicts as unsafe time intervals



Constraints become time intervals, not integer timesteps.  
Still absent: turn radius, actuation lag, localization noise.

CCBS (Andreychuk et al., IJCAI 2019) replaces integer timesteps with continuous time: agents are discs moving along straight edges with arbitrary durations, and conflicts become unsafe time intervals. Turn radius, actuation lag, and disturbance remain outside the model.



## The Gap in Classical MAPF

- **Multi-Agent Path Finding (MAPF)**: move a team of agents to goals without collision. Solvers: **CBS/ECBS** (conflict-based search), **PIBT**, **LaCAM/LaCAM\***, and lifelong **RHCR**.
- All operate on a **discrete grid with simplified kinematics** (unit moves / rotate-in-place) and **no disturbance model**.
- A "conflict" is a shared vertex/edge (or swept-disc overlap) — **dynamics-blind**.
- Consequence: realized collisions on a real floor (turn-radius limits, localization noise, actuation lag) are **out of scope**.

The discrete plan is provably conflict-free *on the graph* — yet the continuous robot can still collide. That is the gap HJ reachability fills.



## The Scaling Anchor: RHCR<sup>1</sup>

- **Rolling-Horizon Collision Resolution:** decompose lifelong MAPF into a sequence of **Windowed MAPF** instances; resolve collisions only within a bounded time window  $w$ , ignore beyond.
- Scales to **1,000 agents** on warehouse maps; co-authored by **J. Durham (Amazon Robotics)**.
- Current frontier: **LaCAM\*** (10,000+ agents, near-optimal, anytime), **PIBT** (thousands in ms), learning-based lifelong (imitation, guidance-graph, MAPF-GPT).
- **All still discrete-grid, deterministic.**

HJ-Gauss does not compete with LaCAM's combinatorial search; it **certifies** the plan under continuous dynamics — a different, complementary axis.

<sup>1</sup> Li et al., AAAI 2021

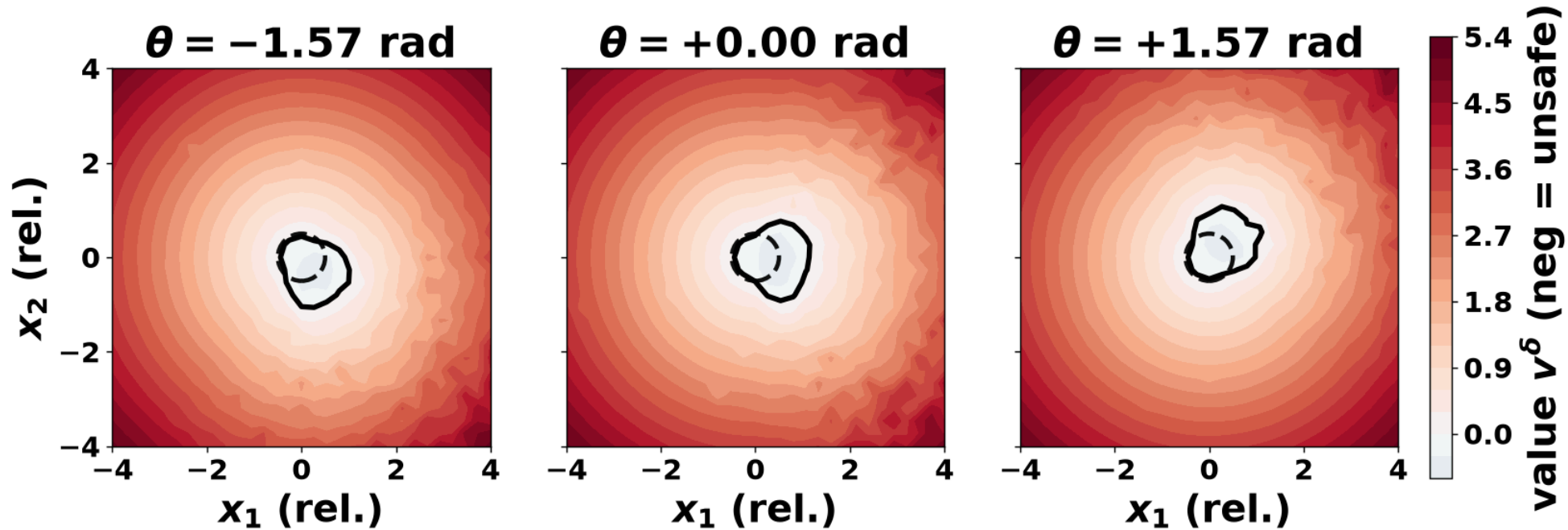
## The Mapping: HJ-Gauss Constructs → MAPF Roles

| HJ-Gauss construct                               | MAPF role                                                                                                |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Windowed BRT</b> over $(0, T]$ , set $T = w$  | Continuous, disturbance-robust replacement for the <b>RHCR rolling window</b> (innovation #1).           |
| <b>Pairwise Dubins BRT membership</b>            | Dynamics-aware <b>conflict predicate</b> replacing the geometric/disc check in CBS/CCBS (innovation #2). |
| Relative-frame tube, computed once               | Reused across all pairs & timesteps $\Rightarrow$ certificate cost <b><math>O(1)</math> in agents</b> .  |
| <b>Value gradient</b> $Dv \propto \nabla \log p$ | A <b>score</b> : certified guidance term for diffusion planners (innovation #3).                         |

The Dubins pursuit-evasion BRT from Part 5 **is** the canonical pairwise MAPF conflict, computed rigorously.

## The Dynamics-Aware Conflict Set (Pairwise Dubins BRT)

Pairwise Dubins BRT (black = inevitable-collision boundary) vs naive disc (dashed)



Black = certified inevitable-collision boundary; dashed = naive disc. The BRT bulges beyond the disc in a **heading-dependent** way — what a dynamics-blind check cannot see.

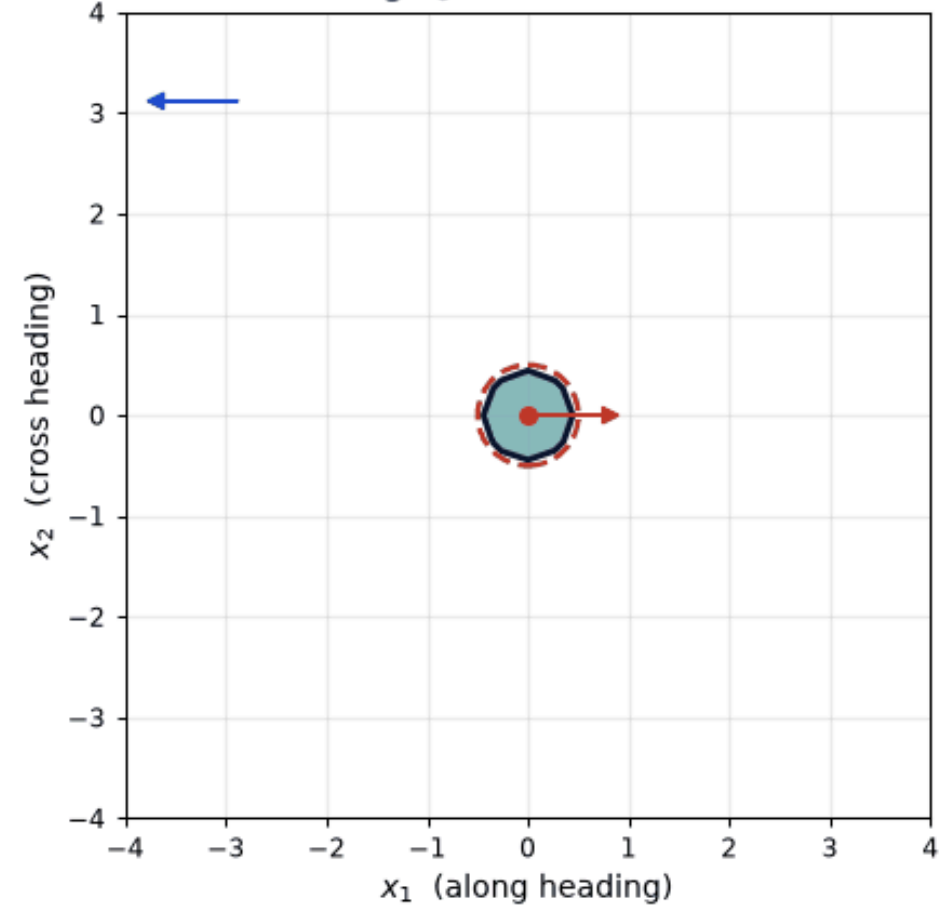
## ⚙ What the Certified Set Looks Like

- Sweeping the relative heading  $x_3$  makes the certified set **rotate and deform**.
- A near head-on geometry grows a long lobe; a near-parallel one stays compact.
- The dashed red circle is the dynamics-blind disc, **fixed for every heading**.

This is the object a swept-disc test approximates with a circle.

Windowed pairwise conflict set,  $w = 0.6$  s

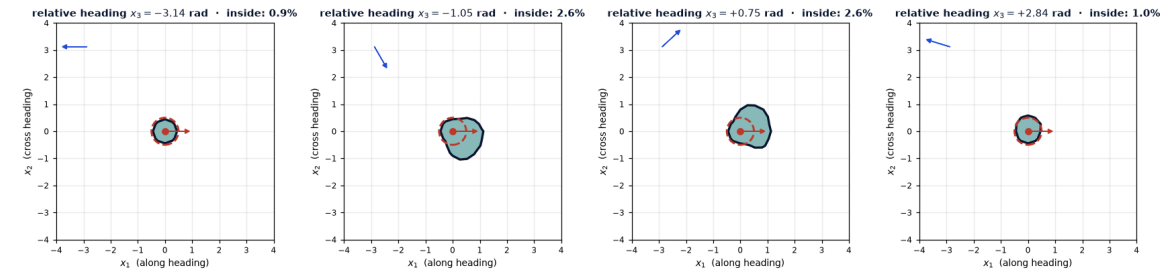
relative heading  $x_3 = -3.14$  rad • inside: 0.9%



teal = certified inevitable collision • dashed red = dynamics-blind disc

## ⚙ Reading the Heading Sweep

- Only about **2%** of the relative-state box is certified inevitable-collision at  $w = 0.6$  s.
- So the gate is **selective**, not blanket-conservative: it fires on geometry that actually traps the pair.
- Where the tube exceeds the disc, a geometric planner is blind; where it falls inside, the disc is needlessly cautious.



## Falsifiable Hypotheses (Stated to Be Proven Wrong)

Naming: **HJ** = Hamilton-Jacobi (the method); **HB- $n$**  = hypothesis  $n$ .

| ID             | Hypothesis                                               | Null we try to reject                  | Metric                       |
|----------------|----------------------------------------------------------|----------------------------------------|------------------------------|
| HB-1           | HJ-shield is <b>sound</b>                                | Admits collisions geometric catches    | Realized collisions          |
| HB-2           | HJ is <b>dynamics-aware</b>                              | Geometric never misses what HJ catches | Collisions, geom vs HJ (CRN) |
| HB-3           | Memory $O(N \cdot n)$ , flat in resolution               | Grows like grid                        | Peak memory                  |
| HB-4           | Certified gate costs <b>little throughput</b>            | Large makespan inflation               | Throughput / flowtime        |
| HB-5           | Fast enough (precompute + $O(1)$ lookup)                 | Too slow for the loop                  | Wall-clock / latency         |
| HB-6 (stretch) | $Dv$ -guidance reduces collisions in a diffusion planner | No reduction                           | Guided vs unguided           |



## The AMFS Pipeline: World and Planner

---

- **World:** an RHCR-style structured warehouse grid — storage blocks, single-cell corridors, border endpoints — with **lifelong** task reassignment.
- **Planner:** windowed prioritized planning (space-time  $A^*$ ), the RHCR rolling horizon, **shared by both policies**.
- **Executor:** a continuous unicycle/Dubins rollout with bounded turn rate and bounded disturbance, where realized collisions are detected in continuous space.

The planner is held fixed on purpose. Only the safety layer varies.



## The AMFS Pipeline: The Two Policies

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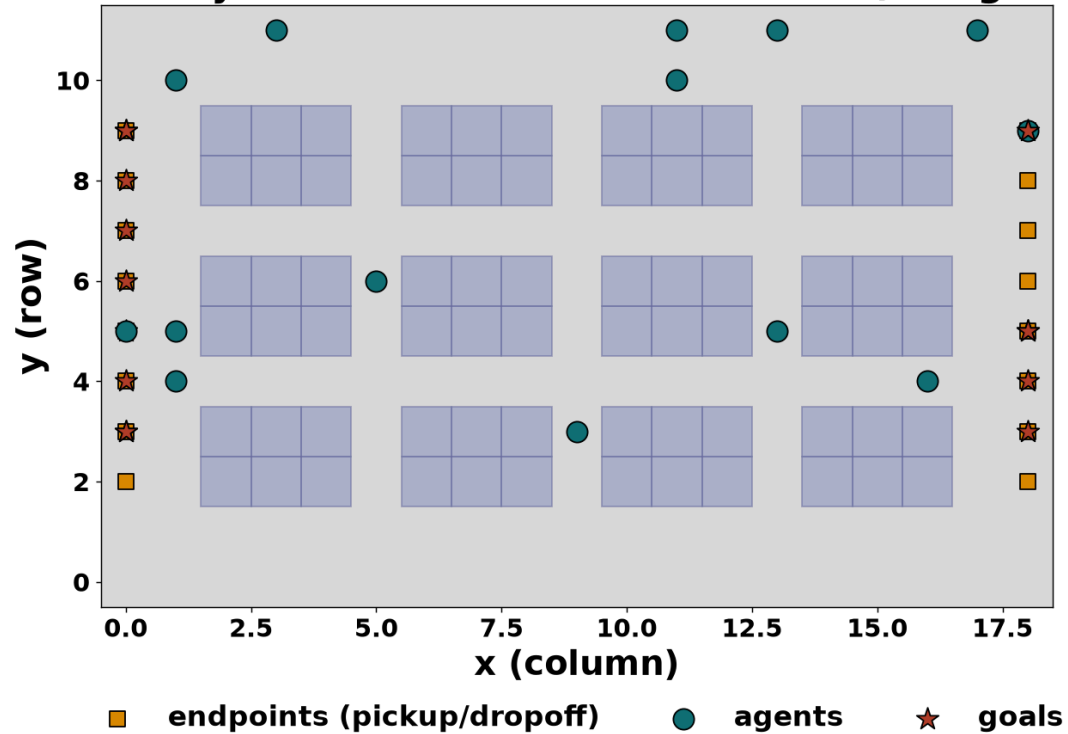
- **Geometric** — the classical dynamics-blind vertex/edge conflict test, unshielded.
- **HJ** — the same planner **plus** the windowed-BRT runtime shield.
- Both run under **Common Random Numbers**: identical spawns, tasks, priorities, and disturbance realizations for a given seed.
- **BRT precompute**: the JAX HJ sampler (Algorithm B) on the Dubins-relative Hamiltonian, cached once; the online loop is numpy-only  $O(1)$  lookups.

One clean contrast: the only difference between the two arms is whether the certificate is actuated.



## The Simulated Floor

**RHCR-style structured warehouse floor (14 agents)**

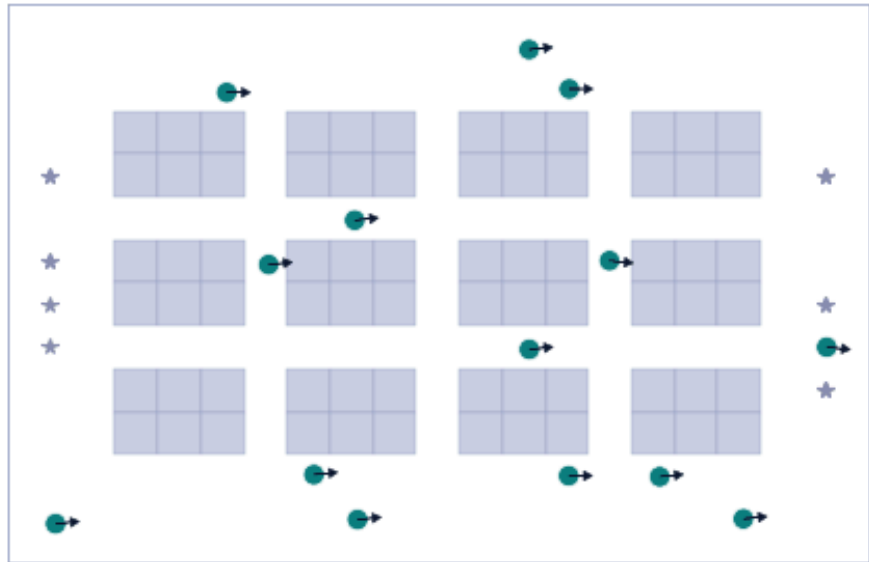


RHCR-style structured floor: storage blocks (obstacles), border endpoints (pickup/dropoff), 14 lifelong agents (circles) heading to goals (stars).

## ▶ Watch It Run: Same Seed, Same Noise, One Difference

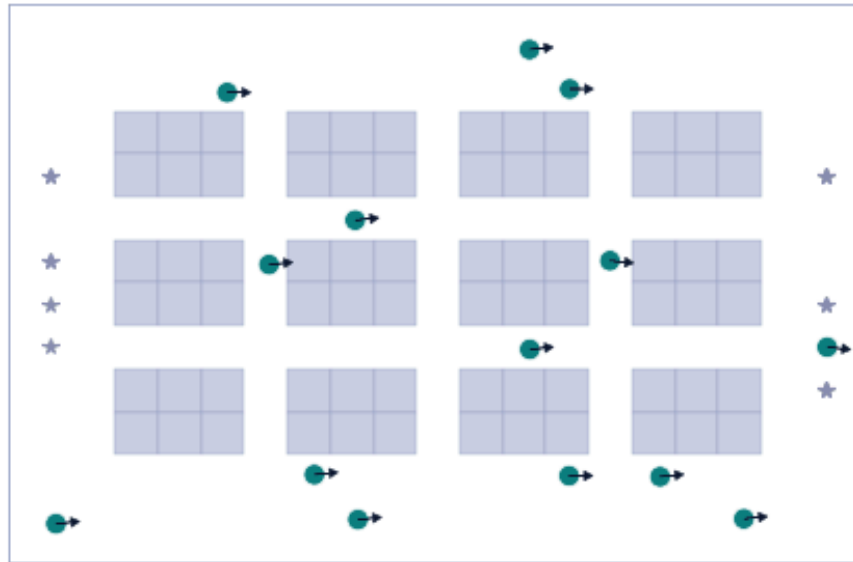
AMFS structured floor · 14 lifelong agents · same seed, same disturbance stream ( $t = 0.0$  s)

Geometric conflict check (dynamics-blind)



realized collisions: 0

HJ-Gauss windowed-BRT shield (certified)

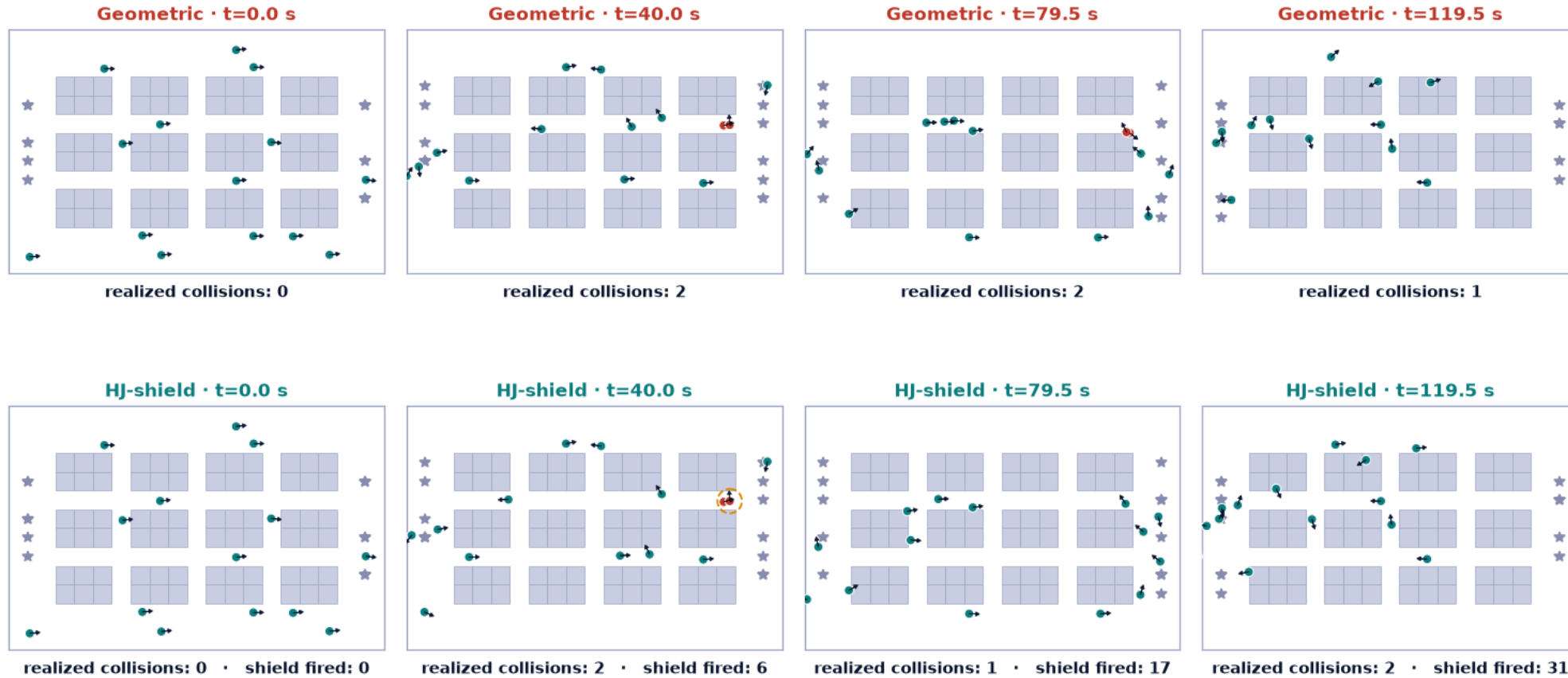


realized collisions: 0 · shield fired: 0

Left: geometric conflict check. Right: the HJ-Gauss windowed-BRT shield. Amber ring = shield firing (agent fleeing), red = realized contact, stars = current goals. Seed 1028 of the 30-seed harness:

98 → 66 collisions at identical throughput (0.825). Animated in the HTML deck; the PDF shows the opening frame.

## ▶ The Same Episode, as a Filmstrip



Four instants from the run above: top row geometric, bottom row HJ-shield. Counters under each panel accumulate realized collisions and shield activations. Identical spawn, task, and disturbance streams throughout (CRN).



## The Key Scientific Finding (Why the Naive Version Fails)

- **Planning-only** BRT predicate made things **worse** (HJ 84 vs geometric 53 collisions in the first trial).  
Two principled reasons:
  - i. the pursuit-evasion BRT is a **worst-case adversarial** set, but agents are **cooperative** — they never apply the BRT's safe control, so plan-time membership does not predict cooperative-tracking collisions;
  - ii. plan-time checks see only **discrete nodes at integer times**; collisions happen in **continuous time under disturbance**.
- **The certificate only bites when *actuated***. Fix: use the windowed BRT as a **runtime shield** — the lower-priority agent in an in-BRT pair **flees** (a braking "spin-in-place" bug had it get hit).

This is the arc: a negative result diagnosed to a principle, then the correct mechanism. Actuation, not inspection.



# The Multiseed Statistical Harness

---

Mirrors the WIP-forecast harness; numpy-only:

- **$\geq 30$  seeds** (configurable);
- **MSER-5** warm-up truncation of the per-tick collision series (+ Welch running-mean cross-check);
- **Bootstrap 95% CIs** on every headline metric;
- **Common Random Numbers (CRN)** across policies (spawn/task/disturbance streams keyed to the seed) for a **paired** comparison;
- **Paired bootstrap CI** for  $\text{mean}(h_j - \text{geom})$  + **Holm-Bonferroni** step-down FWER control across the HB family.

Single-seed results were noisy and even inverted; the harness is what turns a direction into a **defensible** claim.

## Results: 30 CRN-Paired Seeds (14 Agents)

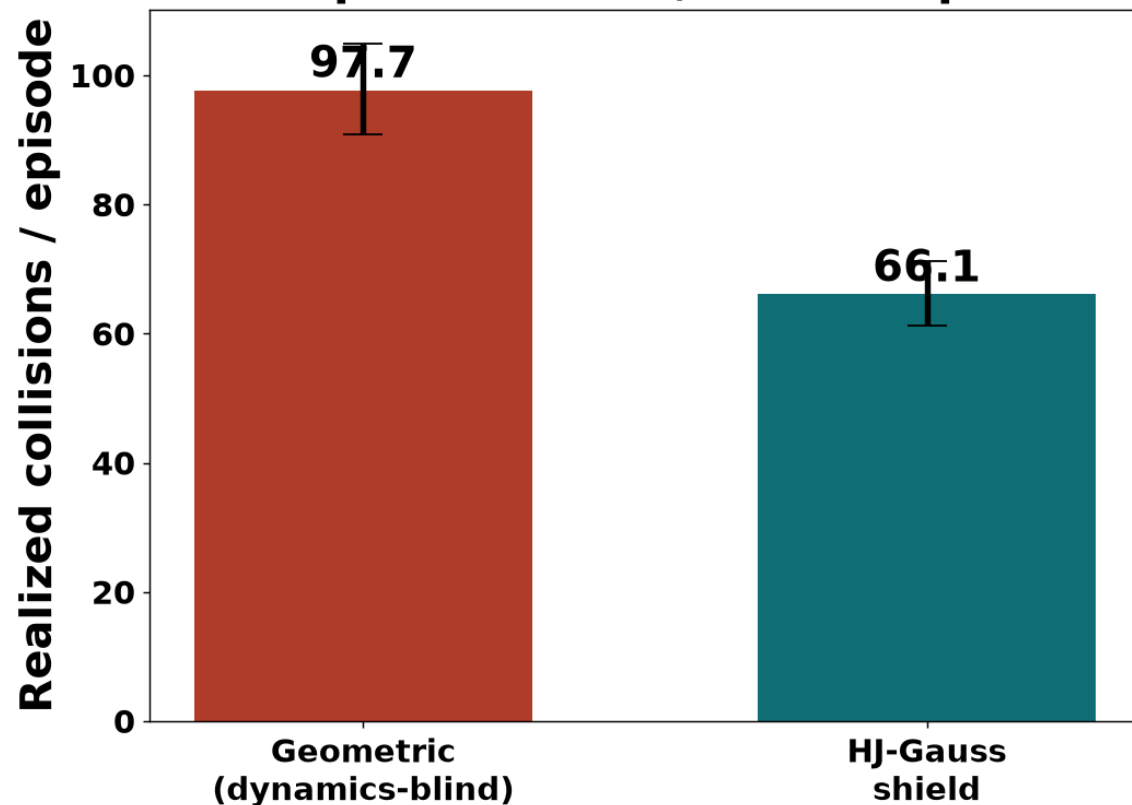
| Metric                  | Geometric          | HJ-shield         | Paired diff hj-geom (Holm)           |
|-------------------------|--------------------|-------------------|--------------------------------------|
| Realized collisions     | 97.7 [90.8, 104.9] | 66.1 [61.2, 71.2] | <b>-31.6 [-38.0, -25.4]</b> , reject |
| Collision rate (MSER-5) | 2.48 /tick         | 1.67 /tick        | <b>-0.81 [-0.98, -0.64]</b> , reject |
| Throughput              | —                  | —                 | 0 [0,0], not rejected                |
| Flowtime                | —                  | —                 | 0 [0,0], not rejected                |
| Shield interventions    | 0                  | ~1024             | (activity, by design)                |

- **~32% fewer realized collisions**, significant under Holm-Bonferroni, at **no throughput/flowtime cost** in this model.
- Aggregated sweeps: 18-37% reduction across densities and disturbance levels.

HB-1/HB-2 rejected in the safe direction; HB-3/HB-4 show no throughput penalty. The certificate, actuated, works.

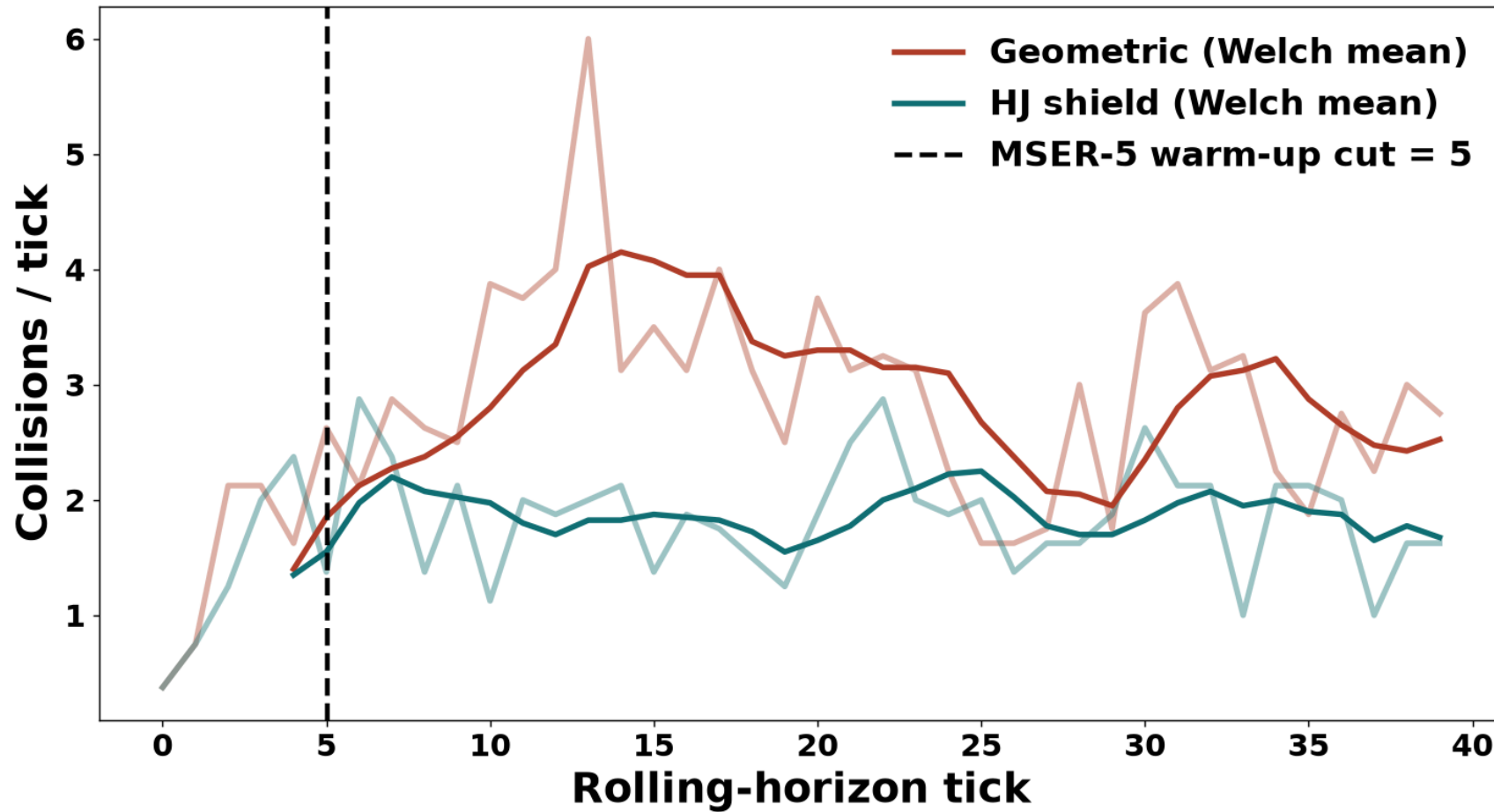
## **Headline: ~32% Fewer Realized Collisions**

**HJ shield cuts realized collisions ~32%**  
**(30 CRN-paired seeds, bootstrap 95% CI)**



30 CRN-paired seeds, bootstrap 95% CI. Geometric 97.7 → HJ-shield 66.1; paired diff **-31.6 [-38.0, -25.4]**, Holm  $p \approx 0$ .

## Per-Tick Collisions & MSER-5 Warm-Up

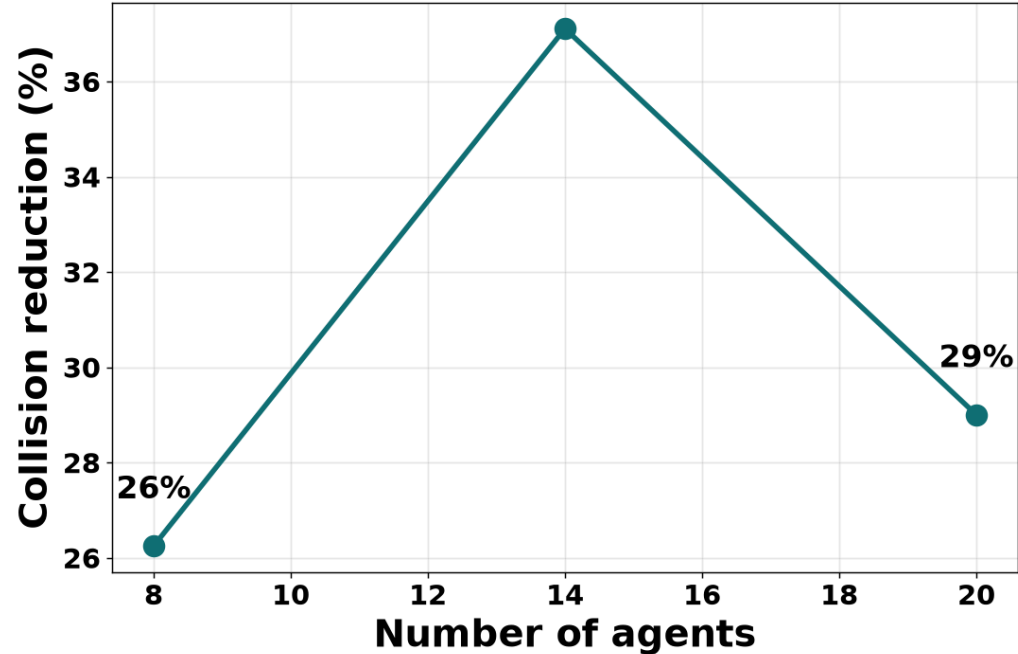


HJ shield (teal) sits below geometric (red) at every steady-state tick; the dashed line is the MSER-5 warm-up cut discarded before averaging.



## The Effect Holds as the Fleet Grows

HJ shield collision reduction vs fleet size (4 seeds/point)



Collision reduction persists across agent densities (8-20 agents) — the certificate's value does not wash out with congestion.



## Verdict Table

| ID   | Prediction               | Result                                                    | Verdict         |
|------|--------------------------|-----------------------------------------------------------|-----------------|
| HB-1 | Shield is sound          | HJ never certifies a colliding plan safe; collisions ↓    | ✓               |
| HB-2 | Dynamics-aware advantage | 31.6 fewer collisions per run, paired, Holm $p \approx 0$ | ✓               |
| HB-3 | $O(N \cdot n)$ memory    | 7.2 MB at $n = 45$ ; flat in resolution                   | ✓ (paper)       |
| HB-4 | Little throughput cost   | Throughput/flowtime diff = 0 in this model                | ✓ (with caveat) |
| HB-5 | Fast enough              | BRT precompute ~10 s; $O(1)$ online lookup                | ✓               |
| HB-6 | Diffusion-guidance helps | Deferred to Option C                                      | ⌚               |

HB-4's zero difference is partly a modeling artifact (next: Dirty Laundry). Everything else holds under the harness.

# Part 7

## The Diffusion Connection

The Cole-Hopf kernel *is* the diffusion kernel.

## C A Mathematical Identity, Not an Analogy

- The Cole-Hopf step turns the HJ PDE into the **heat equation**, whose fundamental solution is the **Gaussian transition kernel** — the identical kernel that **score-based generative models** (DDPM, score matching) and **diffusion policies** use in their forward process.
- Under  $v = -\delta \log \phi$ , the recovered value gradient is a **score function**:

$$Dv \propto \nabla \log \phi = \nabla \log p.$$

- Our Feynman-Kac gradient estimator (weighted-mean shift) is structurally a **training-free score estimator**.

Certified reachability and diffusion trajectory generation **meet at the same Gaussian kernel**. This is not a metaphor; it is the same PDE.

## Consequence: A Certified Guidance Term

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- Diffusion trajectory planners (Diffuser, Decision Diffuser, Motion Planning Diffusion) generate paths by **denoising** but carry **no hard safety guarantee**.
- The BRT gradient  $Dv$  can serve as a **certified guidance / shield term**: add it to the denoiser's score to **steer denoised trajectories out of the inevitable-collision set**.
- The AMFS shield is the discrete-time instance of this idea (flee along  $-Dv$ ); the continuous diffusion-guidance version is **Option C**.

A reachability-certified **shield for neural planners** — the missing safety guarantee in the learning-based MAPF line (imitation, guidance-graph, MAPF-GPT).

## The Careful Claim (for the Q&A Skeptic)

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- $Dv \propto \nabla \log \phi$  is a **genuine score of the Cole-Hopf-transformed value density**  $\phi$ .
- **But**  $\phi$  is **not** the data distribution a diffusion planner learns — the objects coincide *structurally* (same kernel, same  $\nabla \log$  form), not *identically*.
- The rigorous joint-work statement: use the **certified** reachability score as a **guidance/projection** term on the learned generative score, inheriting the BRT's worst-case guarantee as a shield while the learned score supplies task performance.

State it carefully and it is bulletproof; overstate it and a diffusion expert will (rightly) push back.  
Structural identity + guidance role, not "our value function is your data score."

## C Joint-Work Surface with a Diffusion-Planner Team

- **Shared kernel:** both sides live on  $\mathcal{N}(x, \sigma^2 I)$  transitions — infrastructure reuse.
- **Certified shield:** wrap any learned proposer (BC, CQL/IQL, diffusion) with the windowed-BRT flee/guidance term; report blocked actions and throughput margin separately.
- **Score as drift:** the importance-sampling tilt  $\theta^* = -\sigma^2 c Dv$  is literally a score-guided proposal — a learned score could serve as the sampler's drift and vice versa.
- **Benchmark:** extend League-of-Robot-Runners / RHCR with a disturbance model → a **robust-throughput** axis.

One decision contract, two evidence environments: certified reachability and learned generation, meeting at the Gaussian kernel.

# Part 8

## Dirty Laundry

Limits, caveats, and boundaries.



## Theoretical Limits

- **Quasi-linearization residual  $\mathcal{E}_{\text{ql}}$**  floors accuracy for non-quadratic  $H$ ; the scheme is **closed-form only when  $H = \frac{1}{2}|p|^2$** . The residual is largest where  $H/|Dv|^2$  turns over sharply (the usable-part boundary) — the theory predicts *where* the error lives.
- **Worst-case sample complexity is exponential in  $1/\delta$**  (Hoeffding + crude  $\mu$  bound:  $\sim e^{2(g_{\max}-g_{\min})/\delta}$ ), while viscosity error is only  $O(\sqrt{\delta})$  — a genuine bias-variance tension. Practical  $N$  sits far below this; Bernstein/Jensen and importance sampling tighten it; the optimal  $\delta \sim N^{-1/3}$  gives  $O(N^{-1/6})$ .
- **Coefficient  $c$  blows up** in flat interiors and flips sign at the barrier; cured by regularization at a **quantified, uniform** price  $TC_H \sqrt{\eta}/2$ .

None of these are hidden — each has a theorem, a remark, and a mitigation. The certificate is correct by construction (abstention band).



## Reporting & Accuracy Caveats

- The Crandall-Lions  $O(\sqrt{\delta})$  bound and the reported  $L^2 / L^\infty$  errors are in **different currencies** (inviscid-vs-viscous sup-norm vs MC-vs-grid); compared with explicit care.
- **Averaging  $\neq$  agreement**: even the 30-seed-averaged field differs from the grid (paired Wilcoxon,  $p < 10^{-8}$ ) — the systematic residual remains.
- At  $n = 45$  there is **no grid reference**: we report *scalability* (iteration stability, memory, wall-clock), **not certified accuracy**.
- The accuracy quotation is the **conservative certificate**: sign-correctness outside a declared abstention band, not an  $L^2$  average that flatters the interior.

$L^2$  flatters;  $L^\infty$  is dominated by the band we already declare undetermined. We quote the certificate, not the flattering average.



## MAPF-Application Caveats

- **Adversarial vs cooperative:** the BRT is a worst-case set; using it as a cooperative-MAPF predicate is **safe but conservative**, and only reduces realized collisions when **actuated** (shield), not merely checked at plan time — this was our diagnosed negative result.
- **"Free safety" is a modeling artifact:** in the AMFS executor the discrete planner is authoritative for task progress, so shield-induced fleeing does **not** delay arrivals → HB-3/HB-4 diffs are 0. A **physical** executor should let fleeing delay arrivals, introducing a small, real throughput cost.
- **Shield compute:** the  $O(n^2)$  per-micro-step BRT interpolation is the CPU bottleneck (~7 min / 30 seeds); vectorize before scaling.
- **Scope:** Dubins agents, one floor topology, single disturbance level — a PoC, not a fleet study.

The result is real and significant; the *magnitude* and cost need a more physical executor and larger sweeps (Option C, GPU).



## Source & Scope Boundaries

- Grid solvers remain **preferable for  $n \leq 4$**  (smaller constant, no sampling variance); HJ-Gauss is the method of choice only in the high-dimensional regime. The paradigms are **complementary**.
- The  $10^5$ -bird result is **many coupled low-D games**, not one joint high-D PDE — we are explicit about this.
- All experiments are **single-CPU**; GPU/Lambda scaling (Part 9) is future work, not claimed here.
- The diffusion connection is a **structural identity + guidance role**, not a claim that our value density equals a learned data distribution.

Complementary, decoupled, single-CPU, structural — four boundaries stated plainly so no claim is overread.

# Part 9

## Conclusions, Future Work, References

## 🚩 Conclusions

- **HJ-Gauss** turns the nonlinear viscous HJ PDE into a **sequence of linear heat equations** (Cole-Hopf), solved as **Gaussian Feynman-Kac expectations** by a frozen-coefficient **Picard iteration (Algorithm 1)**.
- Memory drops from grid  $O(M^n)$  to  $O(N \cdot n)$  — storage- and discretization-free.
- **Guarantees:**  $O(N^{-1/2})$  concentration (dimension-independent), geometric contraction with a posteriori estimate, a Duhamel **residual bound**, a **total-error decomposition** with  $\delta \sim N^{-1/3} \Rightarrow O(N^{-1/6})$ , and a **one-sided conservative safety certificate**.
- **Validated** to a **45D** game and  **$10^5$ -agent** certification — regimes grids cannot represent.
- **For autonomous mobility:** a dynamics-aware, disturbance-robust **conflict predicate/shield** for MAPF (windowed BRT = RHCR window), and — via the Gaussian/score identity — a **certified guidance term for diffusion planners**.

**Certified safety is the sign of a value function. HJ-Gauss computes that sign where grids cannot — and hands the same object to your learned planners.**

## 🚩 Future Work

- **Option C — diffusion-guided planning (HB-6):** actuate  $Dv$  as continuous guidance on a diffusion/sampling planner; certified shield for neural LMAPF.
- **Physical MAPF executor:** let shield fleeing delay arrivals → measure the *real* throughput-vs-safety frontier (robust-throughput benchmark).
- **GPU / Lambda Labs scaling:** vectorize the shield's BRT interpolation; run the harness as a Metaflow flow (fan-out over configs × seeds); scale the BRT precompute (bigger grids, higher-D dynamics) on H100s.
- **Adaptive importance sampling** in high dimensions; Bernstein-tight sample budgets.
- **Tighter integration** with safe-RL / policy-certification pipelines.

The scaling path ( `imp` /ar-metaflow, W&B, S3 checkpointing) is scoped in `AMFS/infra/lambda_labs_notes.md`.

## References — 1/3

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 Thank You

Questions?

 [ogunmolu@amazon.com](mailto:ogunmolu@amazon.com)

Certified safety is the sign of a value function —  
HJ-Gauss computes that sign where grids cannot.

Backup slides: full proofs (residual Duhamel bound, contraction), murmuration topology theorems, AMFS harness internals, Lambda Labs scaling plan.

# Appendix

## Background & Foundations (Parts A–C)

Hamilton-Jacobi and viscosity theory (A), reachability foundations (B), and the LevelSetPy grid pipeline (C) — reference material for the main talk.



## Who Should Care, and Why

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- **Autonomous mobility / MAPF:** a certified, dynamics-aware conflict predicate that drops into rolling-horizon planners at fleet scale.
- **Safe RL / policy verification:** backward tubes for a closed-loop learned policy are memory-bound on grids;  $O(N \cdot n)$  lifts that bound.
- **Differential games / pursuit-evasion:** the adversarial worst-case reachable set is what HJ-Isaacs computes.

The unifying object is a **value function whose sign certifies safety**. This talk is about computing that sign where grids cannot.



## Who Should Care, and Why

---

- **Differential games / pursuit-evasion:** the adversarial worst-case reachable set is what HJ-Isaacs computes.
- **Diffusion / generative planning:** the Cole-Hopf kernel *is* the score-based diffusion kernel; the value gradient is a score.
- **Air-traffic, collision avoidance, multi-robot control:** the classical application domain of HJ reachability.

The unifying object is a **value function whose sign certifies safety**. This talk is about computing that sign where grids cannot.

# Part A

## Hamilton-Jacobi PDE Theory Recap

Viscosity solutions, vanishing viscosity, and why we need them.



## The Cauchy-Type HJ Equation

Our chief object is the evolution (Cauchy) HJ equation

$$v_t(x, t) + H(t; x, \nabla_x v) = 0 \quad \text{in } \Omega \times (0, T], \quad v(x, 0) = v_0(x) \quad \text{in } \Omega,$$

with  $H : (0, T] \times \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$  continuous and  $g, v_0$  **bounded, uniformly continuous (BUC)**.

- A closely related object is the **scalar conservation law / convection equation**  $v_t + \sum_i f_i(v)_{x_i} = 0$ , whose shock theory motivates the numerics.
- The HJ equation is **first-order, nonlinear, hyperbolic**.

First-order nonlinear hyperbolic PDEs generically fail to have smooth global solutions — the reason the entire viscosity-solution apparatus exists.



## Why Classical Solutions Fail: Crossing Characteristics

- The **method of characteristics** integrates the PDE along curves in  $(x, t)$ .
- For nonlinear  $H$ , characteristics **cross**: multiple characteristics carry conflicting values to the same point  $\rightarrow$  a **shock** / gradient discontinuity.
- Consequence: no global  $C^1$  solution exists in general, **even when  $H$  and  $g$  are smooth**.
- Global analysis via classical PDE theory is "virtually impossible" (Crandall-Lions).

The value function of a differential game develops **kinks and shocks** right at the barrier — the most safety-relevant region. We must define a *weak* solution that survives there.



# Viscosity Solutions<sup>1</sup>

- **Viscosity solutions** are the correct weak solution class for HJ PDEs: they exist almost everywhere, and enjoy **existence, uniqueness, and stability** theorems.
- Definition (sketch) via test functions:  $v$  is a viscosity *subsolution* if for every smooth  $\varphi$  touching  $v$  from above at  $(x_0, t_0)$ ,  $\varphi_t + H(\cdot, \nabla \varphi) \leq 0$ ; *supersolution* with the reverse inequality and touching from below; a **solution** is both.
- This selects the physically-correct kinked solution and discards spurious ones.

Viscosity theory gives us a **unique** object to compute and to certify — indispensable for a safety guarantee.

<sup>1</sup> Crandall & Lions, 1983



## Vanishing Viscosity

Introduce  $\delta > 0$  and regularize (parabolic smoothing, as in gas dynamics):

$$v_t^\delta + H(t; x, \nabla v^\delta) = \frac{\delta}{2} \Delta v^\delta \quad \text{in } \Omega \times (0, T], \quad v^\delta(x, 0) = g(x).$$

- The added Laplacian makes the PDE **parabolic**, hence classically well-posed:  $v^\delta \in C^{2,1}$ , unique, stable, BUC for all  $T$ .
- Traversing the limit  $\delta \rightarrow 0$  recovers the unique viscosity solution.

$\delta$  is the linchpin of HJ-Gauss: it is both the **smoothing** that admits a classical solution *and* the **diffusion coefficient** of the heat equation we are about to expose.



## The Crandall-Lions Error Bound

The viscous solution approximates the inviscid viscosity solution at a known rate:

$$\sup_{t \in (0, T]} \sup_{x \in \mathbb{R}^n} |v(t, x) - v^\delta(t, x)| \leq k\sqrt{\delta}, \quad k > 0.$$

- **Bias scales as  $O(\sqrt{\delta})$ :** smaller  $\delta \rightarrow$  more accurate.
- But (preview) smaller  $\delta \rightarrow$  more concentrated exponential weights  $\rightarrow$  higher Monte-Carlo variance.
- This is the **bias-variance knob** we will formalize in Part 4 ( $\delta \sim N^{-1/3}$ ).

One scalar  $\delta$  trades geometric accuracy against sampling variance. Choosing it well is the practical art of the method.



## Admissible Data on $\mathbb{R}^n$

- Whole-space heat theory requires the datum in  $C(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$ .
- The transformed datum is  $\omega^\delta(0, \cdot) = e^{-c g}$  — **not**  $g$  itself. This matters: the signed distance  $\ell$  is *unbounded above*, but for a bounded target  $g \geq g_{\min}$  and  $g \rightarrow +\infty$  as  $|x| \rightarrow \infty$ , so

$$0 < e^{-c g(y)} \leq e^{-c g_{\min}}, \quad e^{-c g(y)} \rightarrow 0 \text{ as } |y| \rightarrow \infty.$$

- Hence  $e^{-c g}$  is continuous, strictly positive, bounded, decaying  $\rightarrow$  the heat solution is the **unique bounded** one, the integral converges absolutely, and the Gaussian expectation is genuine and **unbiased**.

Only the **lower** bound  $g \geq g_{\min}$  is needed; no upper bound on  $g$  is used. This is what lets us sample over all of  $\mathbb{R}^n$ .

# Part B

## Reachability & Safety Foundations

What a certified safety set *is*, before we compute one.

## What Is Reachability?

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- **Reachability** concerns the *decidability* of a dynamical system's trajectory evolution across a phase space.
- A reachable system is **decidable** when one can compute *all* states reachable from an initial condition **in a finite number of steps**.
- Dual questions:
  - **Forward**: where can the system go from here?
  - **Backward**: from which states is a target inevitably reached (or avoidable)?
- Safety analysis is naturally **backward**: characterize the set of states doomed to enter a danger set, then stay out of it.

Certifying a learned controller, neural policy, or planner means proving it satisfies all specified requirements — a **verification** problem. Reachability is the geometric engine of that verification.

## Backward Reachable Sets and Tubes

- **Backward Reachable Set (BRS)**: states that reach the target at a *specific* time.
- **Backward Reachable Tube (BRT)**: states that reach the target at *some* time in  $[0, T]$  — the safety-relevant object.
- **Reach-Avoid Tube (BRAT)**: states that can reach a goal while avoiding an obstacle set.
- **Robustly-Controlled BRT (RCBRT)**: when the controller must counter a **worst-case disturbance** — the adversarial guarantee.

$$\mathcal{L}([-T, 0], \mathcal{L}_0) = \{x \in \mathbb{R}^n : \exists \beta \in \mathcal{B}(t) \forall u \in \mathcal{U}(t), \exists \bar{\tau} \in [-T, 0], \xi(\bar{\tau}) \in \mathcal{L}_0\}.$$

The strategy structure  $(\exists \beta \forall u)$  encodes the game: disturbance plays a nonanticipative strategy  $\beta$  against control  $u$ .

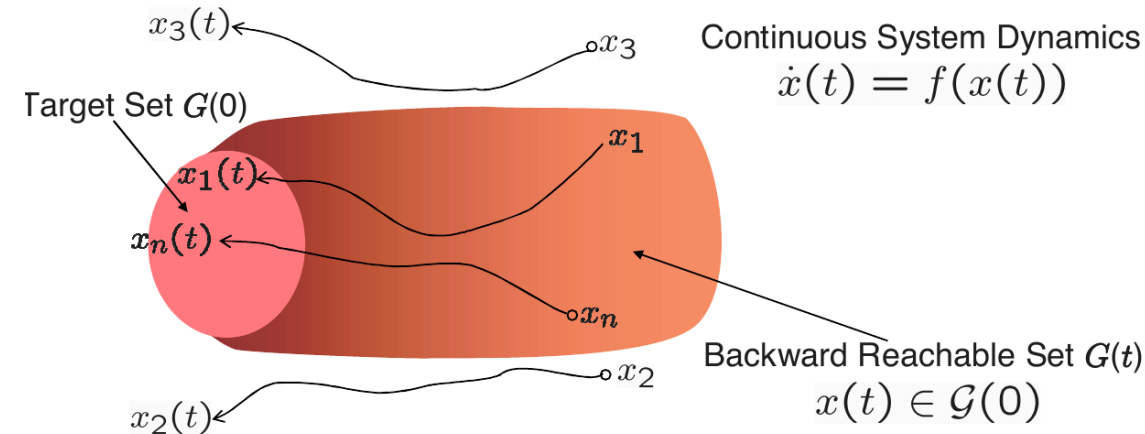


## Why "Backward" Reachable Sets

- A **continuous backward reachable set** is the set of all states from which trajectories can reach a given target set  $G(0)$ .
- Called **"backward"** to distinguish it from the *forward* reachable set.
- **To compute it, run the dynamics backward in time** from the target set.

For safety, the target is usually the *unsafe* set, so the backward reachable set is the set of states doomed to become unsafe.

(Mitchell 2004, Reach Sets and the HJ Equation)



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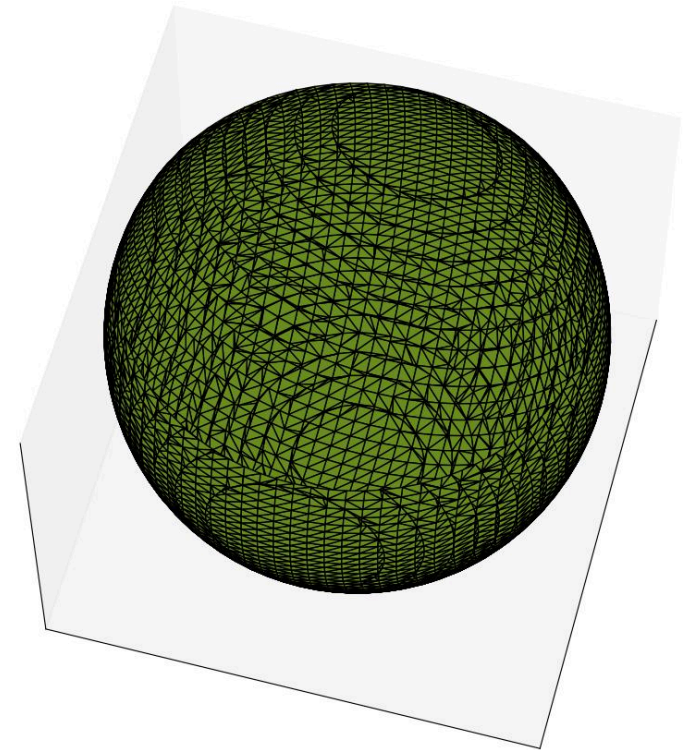
## 🎓 How to *Represent* the Set/Tube?

---

Computing a reachable set poses two coupled problems:

- **Represent** the set of reachable states.
- **Evolve** that set according to the dynamics.

Inset: A sphere as an SDF — LevelSetPy (L. Molu), ACM TOMS 51(2), 2025.

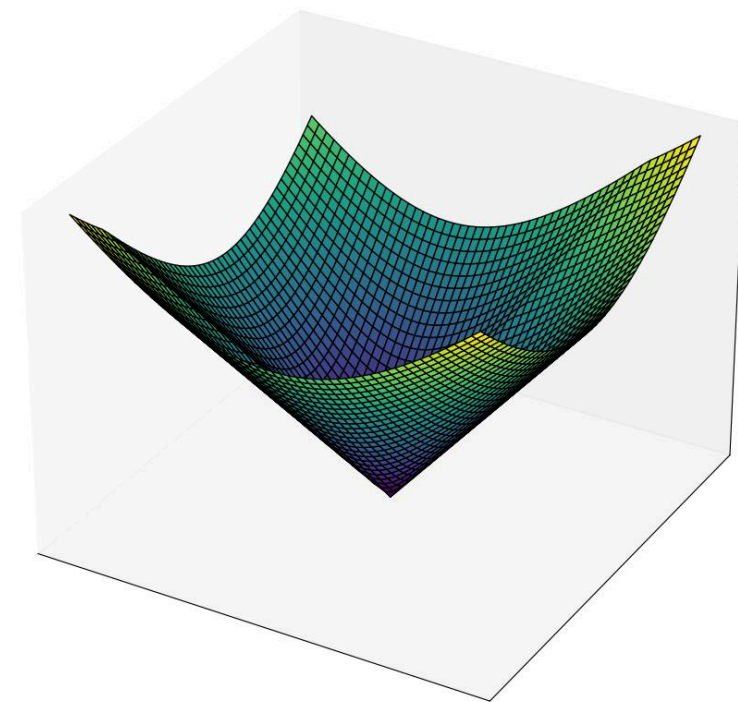


## Implicit Surface Functions

Level-set idea: Represent a set  $G(t)$  as an **isosurface of a scalar function**  $\phi(x, t)$ <sup>1</sup>:

- **State-space dimension does not matter conceptually**: the same  $\phi$  machinery works in any  $n$ .
- Surfaces **automatically merge and separate** as the set evolves.
- **Geometric quantities** (normals, curvature, distance) are easy to compute from  $\phi$ .

Critical talk juncture: the reachable set is the zero sublevel set of a value function  $\phi$ .



Inset: Set union of two spheres — LevelSetPy (L. Molu), ACM TOMS 51(2), 2025. · <sup>1</sup> Osher & Sethian

## How to *Represent* the Set/Tube?

---

For continuous systems  $\dot{x} = f(x)$ , two families:

- **Lagrangian** (forward sets; restricted dynamics/shapes; overapproximation): HyTech, Checkmate,  $d/dt$ , ellipsoidal<sup>1</sup>.
- **Eulerian** (backward sets; general dynamics incl. competitive inputs; **implicit** set representation).
  - **HJ-Gauss** belongs here.

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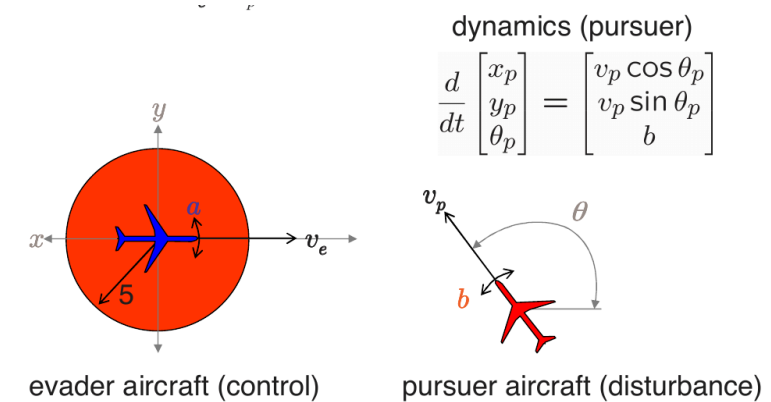
<sup>1</sup> Kurzanski

## Canonical Example: Two Identical Vehicles

Classical collision avoidance:

- Collision if the vehicles come within **5 units** of each other.
- **Evader** picks turn rate  $|a| \leq 1$  to *avoid*;
- **Pursuer** picks  $|b| \leq 1$  to *cause* collision; fixed equal speeds  $v_e = v_p = 5$ .

(Mitchell 2004, Reach Sets and the HJ Equation)



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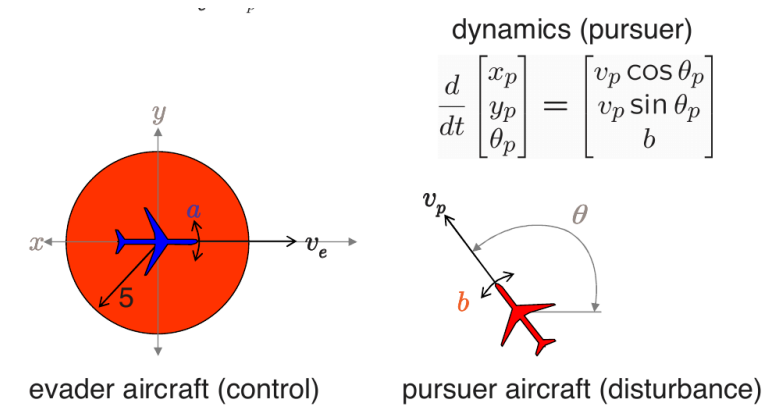
## Canonical Example: Two Identical Vehicles

Classical collision avoidance:

- Work in **relative coordinates** with the evader fixed at the origin.
- State is relative position  $(x, y)$  and relative heading  $\psi$ .

NB: Same relative-coordinate pursuit-evasion (PE) game we solve with HJ-Gauss (Part 5); and the pairwise MAPF conflict of Part 6.

(Mitchell 2004, Reach Sets and the HJ Equation)



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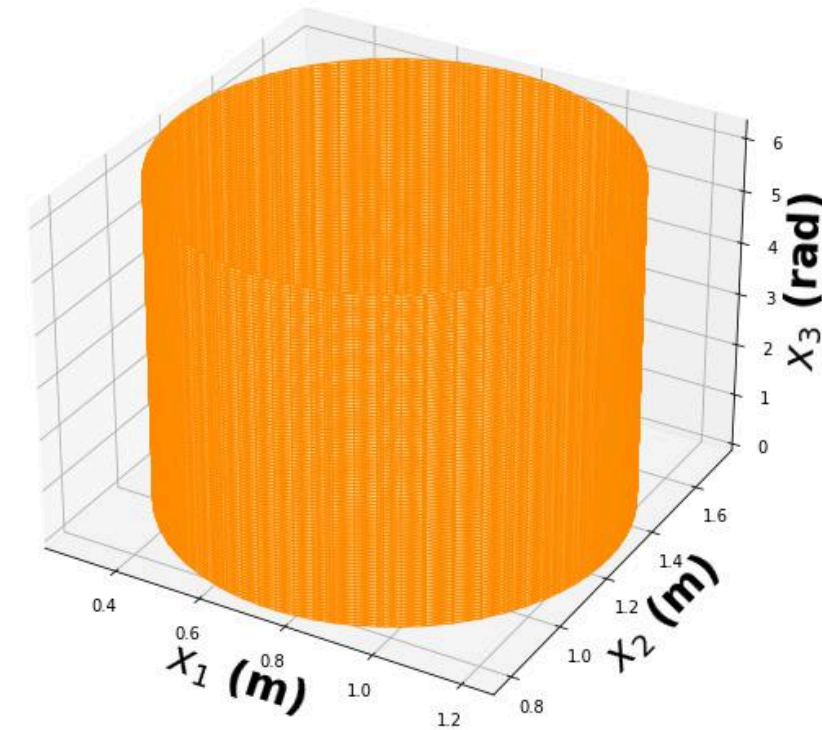
## Evolve: The Time-Dependent HJ Equation

The set evolves by a (modified) Hamilton-Jacobi PDE:

- A **first-order hyperbolic PDE** whose solution can form **kinks** (discontinuous derivatives).
- The right weak solution is the **viscosity solution**<sup>1</sup>.

(Mitchell 2004, Reach Sets and the HJ Equation); viscosity: Crandall-Evans-Lions; level sets: Osher-Sethian · Dubins BRT: LevelSetPy (L. Molu), ACM TOMS 51(2), 2025 · <sup>1</sup> Crandall, Evans, Lions

**BRT at 0.00 secs.**



## Evolve: The Time-Dependent HJ Equation

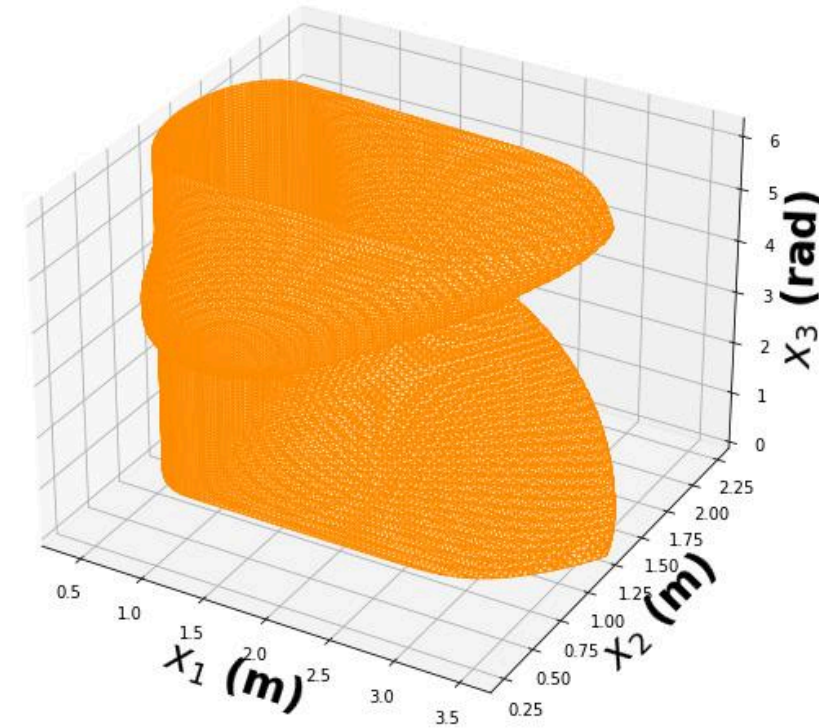
The set evolves by a (modified) Hamilton-Jacobi PDE:

- **Level-set methods** produce convergent numerical schemes<sup>1</sup>:
  - Non-oscillatory high-accuracy spatial derivatives;
  - A stable/consistent numerical Hamiltonian; and
  - Total variation-diminishing (TVD) high-order explicit time integration.

This is the classical grid pipeline of Part C. HJ-Gauss tackles this computational cost.

(Mitchell 2004, Reach Sets and the HJ Equation); viscosity: Crandall-Evans-Lions; level sets: Osher-Sethian · Dubins BRT: LevelSetPy (L. Molu), ACM TOMS 51(2), 2025 · <sup>1</sup> Osher, Sethian

**BRT at 2.50 secs.**



## The Game Value and the Optimal-Stopping Fix

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- The **terminal-cost differential game**:
  - Trajectories  $\xi(\cdot; x, t, a, b)$ ;
  - **Value function**  $\phi(x, t)$  is the viscosity solution of the HJ equation<sup>1</sup>.

This modified/augmented Hamiltonian is the  $\min\{0, \cdot\}$  freeze term on our HJI-RCBRT slide.

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<sup>1</sup> Evans & Souganidis, 1984

## The Game Value and the Optimal-Stopping Fix

---

- To stop trajectories from passing *through* the target  $G(0)$  (so the set is a **tube**, not just a set),
  - **Augment the disturbance input;**
  - The augmented HJ equation solves for the reachable set;
  - The augmented Hamiltonian is the modified  $\min\{0, H\}$  Hamiltonian.

This modified/augmented Hamiltonian is the  $\min\{0, \cdot\}$  freeze term on our HJI-RCBRT slide.

## Three Eulerian Approaches (All Equivalent)

The method sits among Eulerian formulations:

- **Static HJ**: Minimum time-to-reach<sup>1</sup>; (dis)continuous implicit representation; yields optimal-input information.
- **Viability kernels**: Set-valued analysis for very general dynamics<sup>2</sup>; discrete implicit representation; overapproximation guarantee.
- **Time-dependent HJ** (this method): Continuous solution, optimal-input information throughout the state space, high-order accurate.

All three are **theoretically equivalent**; HJ-Gauss is a new *solver* for the time-dependent HJ formulation.

(Mitchell 2004, Reach Sets and the HJ Equation); Falcone/Sethian; Aubin/Saint-Pierre · <sup>1</sup> Falcone; Sethian · <sup>2</sup> Aubin; Saint-Pierre

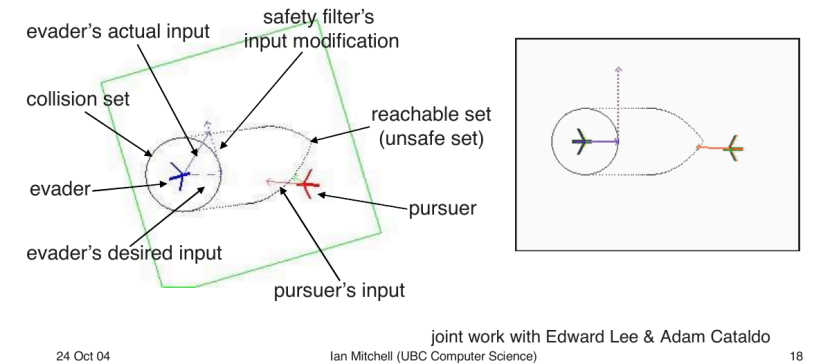
## 🎓 Why It Matters: Two Applications

Reachable sets already drive real-world safety systems:

- **Softwalls for aircraft safety:** Filter evader's input so the pursuer never enters the reachable (unsafe) set — a certified safety filter (with E. Lee & A. Cataldo).
- **Collision alert for ATC:** Flag aircraft pairs whose flight plans intersect and whose relative state enters the collision region.
  - A one-hour Oakland-airspace sample: **1590 pairs, 25 detected conflicts, 2 false alerts.**

These are the ancestors of the certified conflict predicate / shield we bring to warehouse MAPF in Part 6.

(Mitchell 2004, Reach Sets and the HJ Equation); Softwalls with E. Lee & A. Cataldo





## Validation: Merz's Analytic Solution

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- For **identical PE** dynamics, the reachable set admits an **analytic solution**<sup>1</sup>.
- Merz placed the pursuer at the origin; the game is **not symmetric**.
- That analytic solution is used to **validate the numerical algorithm**<sup>2</sup> — the same discipline behind our LevelSetPy-vs-Monte-Carlo comparison in Part 5.

Takeaway of the primer: reachability = represent implicitly, evolve in a HJ PDE game, solve for the viscosity solution.

HJ-Gauss keeps this exactly and changes only *how* the PDE is solved.

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(Mitchell 2004, Reach Sets and the HJ Equation); Merz (1972); Mitchell (2001) · <sup>1</sup> Merz, 1972 · <sup>2</sup> Mitchell, 2001

## Target Set and Zero Level Set

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- The **target set** at horizon  $T$  is the invariant set

$$\mathcal{L}_0(T) = \{x \in \mathbb{R}^n : v(0, x) \leq 0\},$$

robustly controlled over the **distance-to-target** cost  $g(0, x)$ .

- Numerically  $g$  is a **signed-distance function**  $\ell(x)$  whose zero sublevel set is the target: negative inside, positive outside.
- The BRT is recovered as the **zero sublevel set** of the value function:

$$x(t) \in \mathcal{L}(\cdot) \iff v(t, x) \leq 0.$$

## The Hamilton-Jacobi-Isaacs Value Function

The value function of the RCBRT is the **viscosity solution** of the variational HJ-Isaacs equation

$$v_t(t, x) + \min\{0, H(t; x, Dv)\} = 0, \quad v(0, x) = g(0, x),$$

with the **game Hamiltonian**

$$H(t; x, p) = \max_u \min_w \langle p, f(t; x, u, w) \rangle.$$

- The  $\min\{0, \cdot\}$  **freeze** term makes the set only grow (a tube, not a set) — states already captured stay captured.
- For reach-avoid, the variational inequality couples the growth term with the obstacle datum  $\ell$ .

This Hamiltonian is **positively 1-homogeneous in  $p$ , state-dependent, nonconvex, and sign-changing** across the barrier. Those four properties will rule out every convex-duality shortcut.

## Reach vs Reach-Avoid; Sign & Time Conventions

- **Backward-reachability convention:**  $t$  is the **backward horizon** (time-to-go); the datum is posed at  $t = 0$  and the tube grows over  $(0, T]$ .
- **Reach (capture) tube:**  $v_t + \min\{0, H\} = 0$ .
- **Reach-avoid tube (viscous):**

$$\min \left\{ v_t^\delta + H^\delta - \frac{\delta}{2} \Delta v^\delta, \quad g(t, x) - \ell(t, x) \right\} \leq 0.$$

- Physical time is  $t_{\text{phys}} = T - t$ ; the datum is the terminal cost in physical time.

These conventions matter for signs during implementation and for *which* set (reach vs avoid) a negative value certifies. We adopt the backward-reachability viscosity-solution convention throughout.

## Why This Is the Right Safety Object

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- **Worst-case, not average-case:** the RCBRT certifies safety against *all* admissible disturbances — the guarantee a safety case needs.
- **Geometric and set-valued, not trajectory-valued:** it characterizes every unsafe initial condition at once, not one rollout.
- **Composable:** the zero level set can be intersected, unioned, and propagated; it plugs into supervisory control and shielding.
- **Certificate-grade:** a signed value with an error bound yields a decision — SAFE / UNSAFE / UNDETERMINED (Part 4).

The cost of this rigor is computational: solving the HJ(I) PDE. The rest of the talk is about paying that cost at scale.

# Part C

## The Grid Pipeline: LevelSetPy

How Reachable Sets Are Computed Today — and Why It Is  $O(M^n)$ .

Molu. ACM TOMS 2025 · IEEE CDC 2024

TOMS: ACM Transactions on Mathematical Software · CDC: Conference on Decision and Control.

## The Level-Set Idea<sup>1</sup>

---

- Represent the reachable set implicitly as the **zero sublevel set** of  $v(t; x)$  stored on a Cartesian grid.
- Evolve the interface by integrating the HJ PDE on the grid:
  - Discretize space (upwinding),
  - Stabilize the Hamiltonian (Lax-Friedrichs),
  - March in time (TVD-RK).

---

<sup>1</sup> Osher-Sethian

## The Level-Set Idea<sup>1</sup>

- **LevelSetPy** (our prior work) reimplements the 2004 MATLAB Level Set Toolbox in NumPy/CuPy, GPU-accelerated and interoperable with modern Python (PyTorch, SciPy, ROS).

| MATLAB ToolboxLS (2004)   | LevelSetPy (2024/25)         |
|---------------------------|------------------------------|
| CPU-only, single-threaded | NumPy + <b>CuPy GPU</b>      |
| Slow for modern problems  | Fast, batched, portable      |
| No modern-library plug-in | ROS/PyTorch/SciPy compatible |

LevelSetPy makes grid reachability *fast*. It does not change the *memory scaling* — the theme of Part 1.

<sup>1</sup> Osher-Sethian

## Implicit Surfaces & Signed-Distance Initialization

- The target set is initialized as a **signed-distance function**  $\ell(x)$ : negative inside, positive outside,  $|\nabla \ell| = 1$ .
- Stored as an  $n$ -dimensional array over a Cartesian grid:  $M$  points per axis  $\rightarrow M^n$  cells.
- Geometric primitives (spheres, cylinders, half-spaces) and boolean set operations (union/intersection/complement) are implemented as min/max on the level-set field.
- Example: a capture ball of radius  $r$  is  $\ell(x) = \|x\| - r$ ; a cylinder ignores the periodic  $\theta$  axis.

The implicit representation is elegant and closed under set algebra — but it stores the value at **every** grid node.

## Spatial Derivatives via Upwinding

- The co-state  $p = \partial v / \partial x$  must be approximated from grid values with the correct **direction of information flow** (upwinding) to remain stable at shocks.
- **First-order upwinding**: one-sided differences chosen by the sign of the characteristic speed.
- Left/right approximations  $D^-v$ ,  $D^+v$  feed the numerical Hamiltonian; the choice prevents differencing across a discontinuity.

$$v_x(x, t) \approx \frac{\partial v(x, t)}{\partial x} \quad (\text{one-sided, direction by wind}).$$

Naive centered differences ring and go unstable at the barrier; upwinding is the minimal fix, refined next by ENO/WENO.

## ENO: Essentially Non-Oscillatory Reconstruction

---

- **ENO** picks, among candidate stencils, the **smoothest** one to interpolate the derivative — avoiding stencils that straddle a shock.
- Orders implemented: **ENO2** (2nd), **ENO3** (3rd), via `upwind_first_eno2.py`, `upwind_first_eno3.py`.
- Higher order → sharper interface, lower numerical diffusion, at more stencil work per node.

ENO chooses *one* smoothest stencil. WENO improves on this by blending stencils with adaptive weights.



## WENO5: Weighted ENO Shock Capture

- **WENO5**<sup>1</sup> forms a **convex combination** of three candidate substencils rather than choosing one, achieving 5th-order accuracy in smooth regions and non-oscillatory capture at shocks.
- Substencils on the grid index  $i$ :

$$\{i - 3, \dots, i\}, \quad \{i - 2, \dots, i + 1\}, \quad \{i - 1, \dots, i + 3\},$$

combined with nonlinear weights that de-emphasize stencils crossing a discontinuity.

- Implemented in `upwind_first_weno5.py`.

WENO5 is the workhorse for accurate reachable-set boundaries — and its per-node cost is one reason grid solves are expensive even before the memory wall.

<sup>1</sup> Liu, Osher, Chan 1994

## The Lax-Friedrichs Numerical Hamiltonian

- The analytic  $H(x, p)$  is replaced by a **monotone numerical Hamiltonian**  $\hat{H}(x, p^-, p^+)$  using the Lax-Friedrichs flux:

$$\hat{H} = H\left(x, \frac{p^- + p^+}{2}\right) - \frac{1}{2} \alpha \cdot (p^+ - p^-),$$

with dissipation coefficient  $\alpha = \max |\partial H / \partial p|$  over the relevant range (dimension-wise for global LF).

- Monotonicity guarantees convergence to the viscosity solution.

The artificial dissipation  $\alpha$  is a *numerical* cousin of the viscosity  $\delta$  — both stabilize the barrier. HJ-Gauss will trade the grid's LF dissipation for the analytic  $\delta$ .



## Time Integration: TVD Runge-Kutta

Method of lines + **Total-Variation-Diminishing Runge-Kutta**<sup>1</sup>, so the interface does not spuriously oscillate:

- **Forward Euler (1st):**  $v^{n+1} = v^n + \Delta t L(v^n)$ ,  $L = -\hat{H}$ .
- **TVD-RK2:** Euler step to  $v^{(1)}$ , second Euler step, then convex average  $v^{n+1} = \frac{1}{2}v^n + \frac{1}{2}v^{(2)}$ .
- **TVD-RK3:** three stages with convex-combination weights  $(1)$ ,  $(3/4, 1/4)$ ,  $(1/3, 2/3)$ .

TVD-RK preserves monotonicity of the spatial scheme in time. Each stage is a full grid sweep.

<sup>1</sup> Shu-Osher



## The CFL Condition

- Explicit integration is stable only under a **Courant-Friedrichs-Lewy** step restriction:

$$\Delta t \leq \frac{\text{CFL}}{\sum_i \max |\partial H / \partial p_i| / \Delta x_i}, \quad \text{CFL} \in (0, 1).$$

- Numerical information must not travel more than one grid cell per step.
- Finer grids ( $\Delta x \downarrow$ ) force **smaller**  $\Delta t \rightarrow$  more steps  $\rightarrow$  compounding the cost.

CFL couples spatial and temporal resolution: refining space to sharpen the barrier makes *both* the memory and the step count worse.

## Package Anatomy & a Worked Example

| Module                                              | Role                                     |
|-----------------------------------------------------|------------------------------------------|
| <code>grids</code> , <code>initialconditions</code> | Cartesian grids, signed-distance shapes  |
| <code>spatialderivative</code>                      | Upwind first, ENO2/3, WENO5              |
| <code>explicitintegration</code>                    | Lax-Friedrichs Hamiltonians, TVD-RK, CFL |
| <code>dynamicalsystems</code>                       | Vehicle models (Dubins, rockets)         |
| <code>visualization</code>                          | Isosurface / marching cubes rendering    |

- **Worked example:** the rockets-launch pursuit-evasion game; 2D  $(x, z)$  slices of the 3D relative-state BRT, evolved backward over  $(0, T]$ .
- Multi-agent verification: flocks/murmurations partitioned into per-flock games.

A clean, tested, GPU-portable stack — the state of practice we now try to move beyond.

## Grid Complexity: The Accounting

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- **Memory:**  $O(M^n)$  — store the value (and gradient) at every node.
- **Time per step:**  $O(M^n)$  node updates  $\times$  stencil width  $\times$  RK stages, under a CFL-bounded step count.
- GPU acceleration cuts the *constant* and parallelizes node updates, but the **exponent  $n$  is untouched**.

$$n = 6, M = 100 \Rightarrow 10^{12} \text{ cells} \approx 8 \text{ TB per double array.}$$

This is the wall. Part 1 quantifies it and surveys who has tried to climb it.